

Methods and applications for single-cell and spatial multi-omics

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Abstract

The joint analysis of the genome, epigenome, transcriptome, proteome and/or metabolome from single cells is transforming our understanding of cell biology in health and disease. In less than a decade, the field has seen tremendous technological revolutions that enable crucial new insights into the interplay between intracellular and intercellular molecular mechanisms that govern development, physiology and pathogenesis. In this Review, we highlight advances in the fast-developing field of single-cell and spatial multi-omics technologies (also known as multimodal omics approaches), and the computational strategies needed to integrate information across these molecular layers. We demonstrate their impact on fundamental cell biology and translational research, discuss current challenges and provide an outlook to the future.

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Introduction

Humans and many other Eukaryota are composed of billions of cells, belonging to vastly heterogeneous cell types and functional cell states determined by both cell-intrinsic and cell-extrinsic factors. Intrinsically, there is a complex interactive molecular hierarchy of the different 'omics' layers within a cell: from genome and epigenome to transcriptome, proteome and metabolome, and back. Extrinsically, the functional state of a cell can be modulated by its neighbouring cells through direct physical interaction (such as receptor–ligand interactions), through signalling molecules secreted by one cell that can act through receptors on remote cells (such as morphogen signalling pathways), or by other microenvironmental factors (such as chemical compound gradients)¹. Consequently, investigating how multicellular organisms develop from a totipotent single cell and subsequently function, age and develop disease, necessitates single-cell and spatial multi-omics approaches (also known as multimodal omics approaches).

Robust technologies for unimodal (mono-omics) measurements of individual cells, such as single-cell RNA sequencing (scRNA-seq) methods², have already evolved to revolutionize the discovery and understanding of cell types as well as their different functional cell states, cell plasticity upon exposure to external stimuli and drugs, and cell differentiation or reprogramming trajectories³. The power of these technologies is underscored by the instigation of the Human Cell Atlas (HCA)^{1,4} and other consortium-based resources afterwards, which are primarily aimed at creating cellular reference maps of organisms, including the position, function and characteristics of every cell type. However, to develop fundamental understanding of the molecular hierarchy from genome to phenome in individual cells, multi-omics methodologies at single-cell and spatial resolution are necessary. They enable investigation of the intermolecular dynamics between gene regulation on the epigenome level and gene expression on the transcriptome and/or proteome levels unambiguously in the same single cells across development, ageing and disease. Additionally, these technologies enable investigating the impact of acquired genetic variation in the genome of single cells on their own function and phenotypic features, as well as (surrounding) tissue function and more, as exemplified throughout this Review. In recent years, this field has advanced phenomenally and is rapidly maturing both technologically and computationally, enabling broad applications to understand cell biology (Fig. 1).

In this Review, we discuss the fundamental technological and computational principles, state of the art and applicative value of modern single-cell and spatial multi-omics. We focus on methods that provide a comprehensive 'omics' view of at least one molecular analyte, and for single-cell multi-omics are based on next-generation sequencing (NGS), whereas for spatial multi-omics we highlight both NGS-based and imaging-based methods. Although many algorithms have been tailored to analyse the individual molecular layers, here we primarily focus on computational techniques for the integration of information across the different data modalities to maximally leverage the potential of these multi-omics technologies. We end with an outlook to the future.

Single-cell multi-omics methods

Multi-ome measurements from single cells are enabled by different methodological approaches, which can be categorized according to whether the distinct molecular analytes are uncoupled before, during or after sequencing library preparation. As described in Fig. 2 and further below for specific methods, each of these different principles comes with specific advantages and limitations.

Single-cell genomics-plus-transcriptomics

Soon after the establishment of single-cell DNA sequencing (scDNA-seq) and single-cell RNA sequencing (scRNA-seq) protocols^{2,5}, methods for genome-plus-transcriptome sequencing of individual cells were developed (Fig. 1) that rely on one of the four basic principles for multi-ome analysis (Fig. 2).

A first set of methods applies physical separation of DNA and RNA before sequencing library preparation (Fig. 2a). In G&T-seq^{6,7} (Supplementary Fig. 1a), oligo-dT bead-mediated precipitation of polyadenylated (poly(A)) RNA molecules enables their physical separation from the remaining molecules in the cell's lysate, including the nuclear and mitochondrial DNA, either manually or robotically. Captured transcripts are then on-bead primed for full-length cDNA amplification through reverse transcription (RT), template switching and PCR using a Smart-seq2-like reaction⁸, allowing for both short-read and long-read sequencing⁷. Long-read sequencing is preferred for transcript isoform detection. The genomic DNA (gDNA) in the collected supernatant is subjected to whole-genome amplification (WGA) using a method of choice: multiple displacement amplification (MDA), PCR or displacement preamplification followed by PCR (DA-PCR). Single-cell transcriptogenomics applies a similar principle as G&T-seq and is compatible with targeted DNA exome sequencing following MDA⁹. The freedom of choice for downstream processing of the separated poly(A) RNA and gDNA presents a major advantage. Indeed, WGA is not error-free, and different WGA methods present different biases, making some more suitable for the detection of specific classes of genetic variants, as reviewed previously^{10,11}. Similarly, different scRNA-seq methodologies present different performances in sensitivity, transcript coverage and throughput¹².

As alternatives to oligo-dT bead-based separation of poly(A) RNA and gDNA, multiple methods partition cytosolic RNA from nuclear DNA using a two-step cell lysis (Fig. 2b). In SIDR-seq¹³ (Supplementary Fig. 1b), single cells are first subjected to hypotonic lysis, enabling the nucleus to be isolated from the supernatant containing RNA using antibody-conjugated magnetic microbeads. Then, the nucleus is lysed and subjected to MDA-based scDNA-seq, and the supernatant RNA is subjected to Smart-seq2-based scRNA-seq. In DNTR-seq¹⁴ (Supplementary Fig. 1c), the nuclei of single cells are precipitated by centrifugation after cell membrane lysis. While the supernatant cytosol is isolated for Smart-seq2-like scRNA-seq, the nuclear DNA is subjected to direct fragmentation. The latter enables direct PCR-based library preparation for scDNA-seq, thereby circumventing the classic approach of WGA before scDNA-seq library preparation and in part the resulting artefacts associated with it^{14,15}. However, these nuclear–cytosolic partitioning methods are less amenable to comprehensive characterization of mitochondrial DNA and nuclear RNA, and they are confined to the use of intact cells as input. Furthermore, plate-based assays such as G&T-seq, transcriptogenomics, SIDR-seq and DNTR-seq are inherently low throughput and, despite (partial) automation on liquid-handling robotics, still require up to a few days' time⁶. To circumvent this, integrated on-chip microfluidics approaches have been devised that enable separation and parallel processing of cytosolic RNA and nuclear DNA^{16,17}. Such systems largely avoid the introduction of operator bias, can miniaturize the reactions and hence reduce reagent costs, and have the capacity to increase throughput, although this remains to be shown.

Another set of methods relies on a preamplification-and-split approach for uncoupling DNA and RNA molecular analytes (Fig. 2c). In DR-seq¹⁸ (Supplementary Fig. 1d), poly(A) RNA-derived first-strand cDNA is subjected to quasilinear amplification together with the gDNA

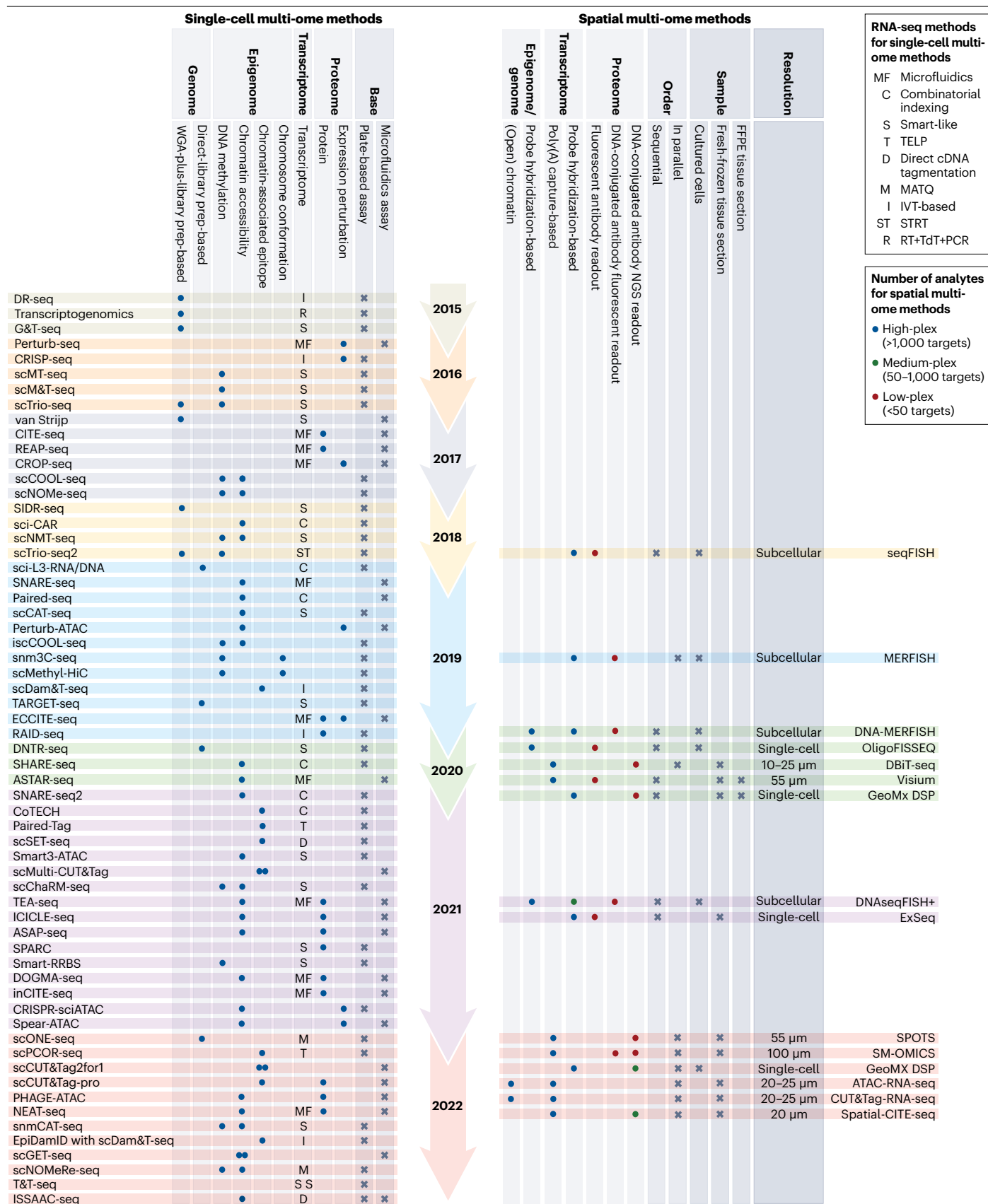


Fig. 1 | Timeline of single-cell and spatial multimodal methods. In addition to the year of publication, other key features of the methods are indicated. For single-cell multi-omics assays, this includes the nature of molecular analytes they analyse as well as the method used for cell barcoding. For spatial multi-omics assays, this includes the resolution, sample type, order and number of analytes

that can be profiled simultaneously. FFPE, formalin-fixed paraffin-embedded; IVT, in vitro transcription; MATQ, multiple annealing and dC-tailing-based quantitative single-cell RNA-seq; STRT: single-cell tagged reverse transcription sequencing; RT, reverse transcription; TdT, terminal deoxynucleotidyl transferase; TELP, tailing extension ligation and PCR.

in a single tube. This reaction is then split, with both aliquots containing preamplified cDNA and gDNA. In one, the preamplified gDNA is further PCR-amplified and converted to a sequencing library, with the caveat that contaminating cDNA will be co-amplified and sequenced into reads indistinguishable from the gDNA sequences. In the sister reaction, the cDNA is further amplified using in vitro transcription (IVT), followed by RT and PCR. Because only the cDNA is tagged with T7 promoter sequence for IVT, gDNA will not be co-amplified, resulting in a 3'-end scRNA-seq library. In the more recently developed TARGET-seq¹⁹, a method for targeted mutation detection and parallel transcriptome characterization of the same single cells, poly(A) cDNA is co-amplified with a mutation of interest targeted by cDNA primers and gDNA primers in a single reaction. This pot is then split to prepare the genotyping and transcriptome sequencing libraries separately. Although DR-seq and TARGET-seq minimize the risk of losing (deoxy)ribonucleic acids in comparison to above-described methods relying on physical separation of DNA and RNA, they can suffer from RNA-derived reads contaminating the gDNA analysis, and are also plate-based and low throughput.

Instead, scONE-seq²⁰ (Supplementary Fig. 1e) follows the principle of seq-split (Fig. 2d) and differentially barcodes gDNA and RNA through, respectively, tagmentation with a 6-nucleotide DNA-barcode-containing adaptor and reverse transcription with a 6-nucleotide RNA-barcode-containing RT-primer. Differentially labelled gDNA and cDNA is then co-amplified and converted to a sequencing library in a single-tube reaction. Following NGS, gDNA-derived and RNA-derived reads are distinguished by their barcode sequence. This methodology hampers sequencing RNA-seq and DNA-seq libraries separately to optimal depths and is plate-based and low throughput, but indicates that relatively simple one-tube reactions are possible for multi-omics measurements of single cells.

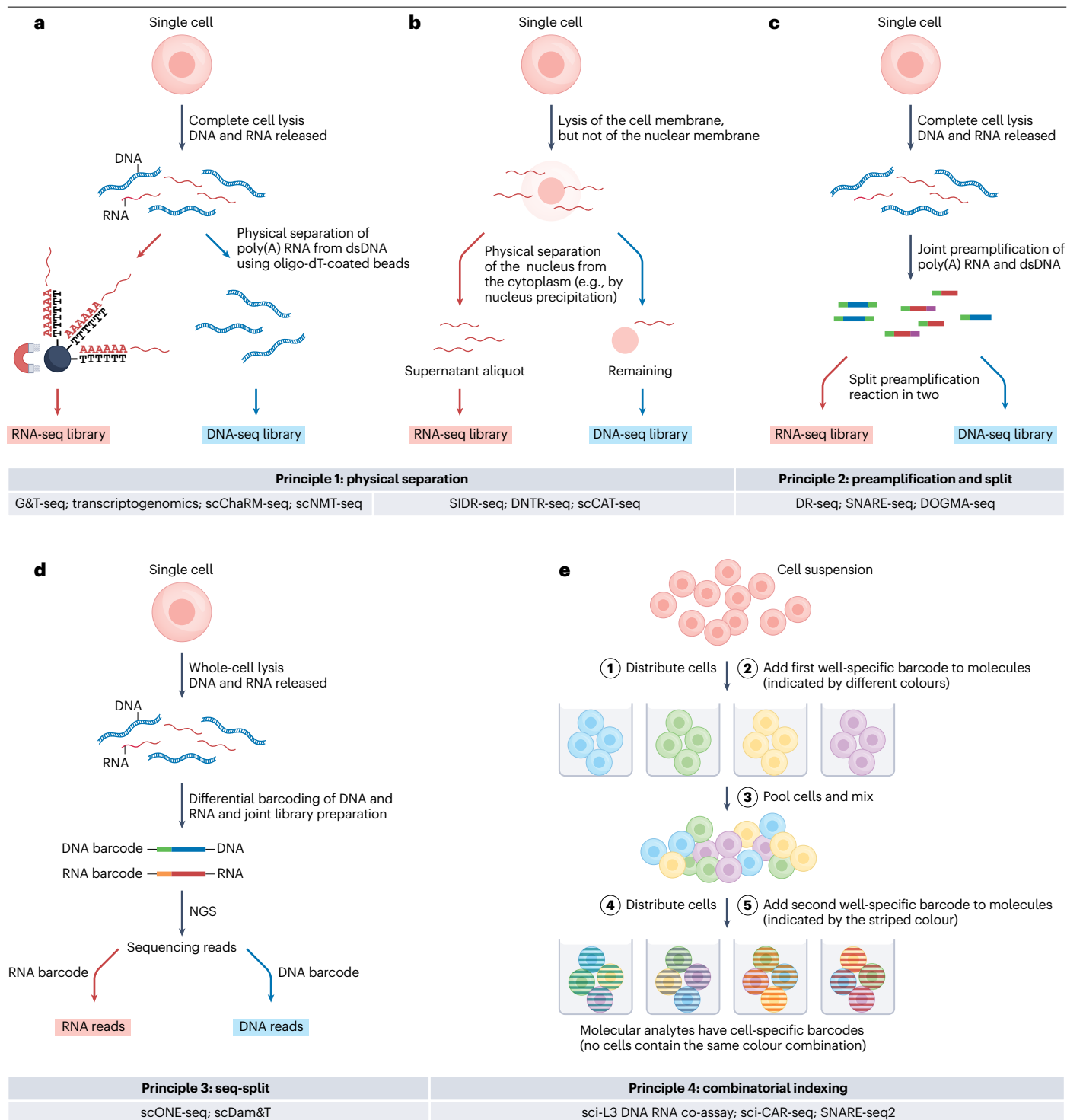
Recently, a highly scalable plate-based technology was established based on the principle of single-cell combinatorial indexing (Fig. 2e) with a three-level indexing scheme and combined with linear IVT-based amplification of the cells' genome and transcriptome (Supplementary Fig. 1f). This sci-L3-RNA/DNA co-assay enables analysis of at least tens of thousands of single nuclei, with the possibility to increase the throughput to more than 1 million cells profiled per experiment²¹. Proof of concept of sci-L3-RNA/DNA was presented on mixtures of male mouse and female human cell lines, proving that the single-cell transcriptomes were organized into the two expected cell clusters and that the matching single-cell genomes were of the correct sex, although high-sensitivity and high-resolution profiling per cell was not demonstrated.

The study of both the genome and the transcriptome of the same cell enables one to unambiguously investigate the impact of acquired DNA mutations, such as DNA copy number aberrations, on gene expression in the same cell. This has important applications for understanding intratumoural heterogeneity, enabling the investigation of the development of different phenotypic cancer cell states among the different genetic subclones that arise, or even within a single genetic subclone. For instance, DNTR-seq identified minor subclones having genetic copy number alterations with associated transcriptional perturbations in paediatric acute lymphoblastic leukaemia¹⁴. Additionally,

transcriptional signatures of the WNT pathway activation learned from scRNA-seq could be explained by mutations detected in the scDNA-seq data from the same cells¹⁷. Furthermore, using tumour model systems exposed to treatment, or direct longitudinal sampling of patient tumour specimens before and during treatment, and analysing them by single-cell genome-plus-transcriptome sequencing will allow investigation of which genetic subclones are more fit to tolerate the drug selection. Additionally, it will allow the study of how cells within these genetic subclones putatively apply cell plasticity to change their gene expression repertoire and accommodate different phenotypic cancer cell states able to withstand drug treatment and, eventually, acquire resistance²². In turn, these approaches might enable the identification of potential cancer cell vulnerabilities, such as druggable molecular players involved in the acquisition of drug tolerance.

Beyond the field of oncology, these multi-omics methods are important for understanding the pathogenesis of other disorders in which somatic genetic variation plays a putative role, including in neurological disorders such as Alzheimer disease, Parkinson disease and others^{23–25}. With the recent discovery that normal tissues are also subjected to an extraordinary amount of mutation, technologies enabling the analysis of the genome and transcriptome of the same single cells will be important to study the role or impact of acquired mutations on phenotypic and functional cellular states, and how these in turn impact development²⁶, cellular competition between normal and genetically aberrant cells²⁷, tissue homeostasis, normal phenotypic variation and ageing²⁸. The power of multi-omics methods in this respect is illustrated by the detection of different transcriptional responses to acquired DNA copy number aberrations^{6,7,14,18} and other forms of mutations⁹. For instance, it was shown using DNTR-seq that structural DNA imbalances lead to both linear and nonlinear transcriptional dosage effects, whereby several genes important for cancer cell growth, such as *MYC* and *TCF7L2*, demonstrated strong dosage compensation and were shown to be mostly unaffected by copy number alterations¹⁴.

Single-cell genome-plus-transcriptome sequencing is also a valuable tool to study the efficacy and safety of genome editing in germline therapy. CRISPR–Cas9 genome editing has potential as a therapeutic tool for the correction of disease-causing mutations. Genome editing of human embryos or germ cells provides the means for introducing heritable genetic alterations, which may reduce the burden of genetic disease in specific familial situations²⁹. Its use is currently a hot topic of international debate around ethics, safety and efficacy. Single-cell genome-plus-transcriptome sequencing approaches will be pivotal to assess on-target and off-target genome edits plus cell phenotypic consequences, as only a few cells are available for analysis from the treated and subsequently in vitro-cultured human embryo. Recently, *OCT4* (also known as *POU5F1*) CRISPR–Cas9-targeted and control human preimplantation embryos were investigated using single-cell G&T-seq^{30,31} as well as single-cell or low-input mono-omics DNA-seq. Regions of loss of heterozygosity in genome-edited cells that spanned beyond the *OCT4* on-target locus, as well as segmental loss and gain of the *OCT4*-containing chromosome 6, were detected in the genomic data, which collectively resulted in unintended genome edits being



identified in ~16% of the human embryo cells analysed. The transcriptome data suggested that the loss of heterozygosity does not lead to the misexpression of other genes adjacent to the *OCT4* locus.

Furthermore, genome-plus-transcriptome sequencing approaches allow genetic variation detected in DNA sequences to be confirmed in the RNA sequences of the same cell, increasing the reliability of the genotyping call^{7,9,13}. This principle has been shown for the

detection of single nucleotide variants contained within expressed genes, forms of structural variation resulting in the expression of fusion genes, and copy number variants resulting in gene expression dosage effects. The genomic and mitochondrial DNA variants detected by genome-plus-transcriptome sequencing approaches, considering potential imperfections¹⁵, can furthermore be leveraged for the construction of a genetic lineage tree of the cells, which can

Fig. 2 | The four general principles for multi-ome measurements from single cells. All principles are visualized with RNA and DNA as example analytes. Principle 1 is based on physical separation of the distinct molecular analytes (parts **a, b**). **a**, Following complete lysis of the isolated cell or nucleus, poly(A) RNA hybridizes to oligo-dT-coated paramagnetic beads, and following magnetic pull-down, the supernatant that contains the genomic DNA is transferred to a new reaction vessel⁷⁹. Alternatively, biotinylated nucleotides are incorporated into RNA-derived cDNA, allowing their capture with streptavidin-coated paramagnetic beads (not shown)⁴². Advantages include flexibility in downstream processing of DNA and/or RNA and compatibility with intact cells and nuclei from fresh and frozen tissue. Disadvantages include potential loss of RNA and/or DNA molecules during physical separation. **b**, In an alternative approach, lysis conditions that rupture the cell but not the nuclear envelope allow separation of nuclear from cytoplasmic molecular analytes, either by precipitating the nucleus with centrifugation or magnetic pull-down followed by aspiration of the cytosolic supernatant^{13,14} or by microfluidic-controlled nucleus-from-cytoplasm separation^{16,17}. Advantages include flexibility in downstream processing of DNA and RNA and availability of non-poly(A) RNA. Disadvantages include loss of nuclear RNA plus some cytoplasmic RNA during nuclear–cytoplasmic separation, loss of mitochondrial analytes, need for intact single cells, incompatibility with frozen tissue and likely incompatibility with mitotic cells (in which the nuclear envelope disaggregates). **c**, In principle 2, termed preamplification and split, distinct analytes are differentially tagged and jointly preamplified, followed by splitting the preamplification reaction for

analyte-specific sequencing library preparations¹⁸. Advantages include minimal risk of analyte loss and compatibility with intact cells and nuclei from fresh and frozen tissue. Disadvantages include limited flexibility, as the preamplification protocol needs to be suitable for all analytes of interest, and risk of cross-contaminating molecular analytes. **d**, Principle 3, termed seq-split, involves analyte-specific barcoding and sequencing library preparation in a single-pot reaction²⁰. Multi-omic information is uncoupled computationally following sequencing. Advantages include minimal risk of analyte loss and compatibility with intact cells and nuclei from fresh and frozen tissue. Disadvantages include that libraries cannot be sequenced separately to optimal depth for each modality, and potential risk of cross-contaminating molecular analytes. **e**, In principle 4, termed combinatorial indexing, molecular analytes of single cells are tagged without isolating single cells²¹. Multiple cells or nuclei are deposited per well of a multi-well plate, whereby each cell or nucleus serves as a reaction container. Each receives an analyte-specific tag and a well-specific barcode. By pooling, mixing and randomly re-distributing the cells or nuclei in subsequent rounds of well-specific barcoding, molecular analytes uniquely barcoded per cell are obtained. Combination with concepts of principle 2 and/or 3 achieves single-cell or single-nucleus multi-omics. Advantages include that there is no need for isolating single cells, ability to achieve extremely high throughput, and compatibility with intact cells and nuclei from fresh and frozen tissue. Disadvantages include typically lower sensitivity, risk of analyte loss and limited flexibility in whole-genome and whole-transcriptome amplification protocols. dsDNA, double-stranded DNA; NGS, next generation sequencing; poly(A), polyadenylated.

be annotated with cell type and functional phenotypic states of cells using the RNA sequencing data of the same cells. As an alternative to analysing naturally occurring somatic mutations, high-throughput methods relying on CRISPR-scarring are available in model systems³², which through recent improvements may enable accurate lineage recording as well as the capturing of ancestral transcriptional states^{33,34}. In combination with scRNA-seq readouts, these promise to revolutionize our understanding of cellular differentiation trajectories.

Single-cell epigenomics-plus-transcriptomics

Advances in the past few years have also moved the mark for methods that can profile a cell's epigenome and transcriptome in parallel (Fig. 1). As with methods analysing the genome and transcriptome, their original designs relied on the physical separation of RNA from DNA or the nucleus, and the subsequent profiling of these separated fractions. More recently, methods for the differential marking of RNA and epigenetic information encoded in DNA have been described, relying on the separation of reads originating from both through restriction digestion, PCR or molecular barcoding. These later methods do not require upfront separation and are therefore more readily parallelized in higher throughput, via preamplification and split, seq-split or combinatorial indexing principles (Figs. 2 and 3).

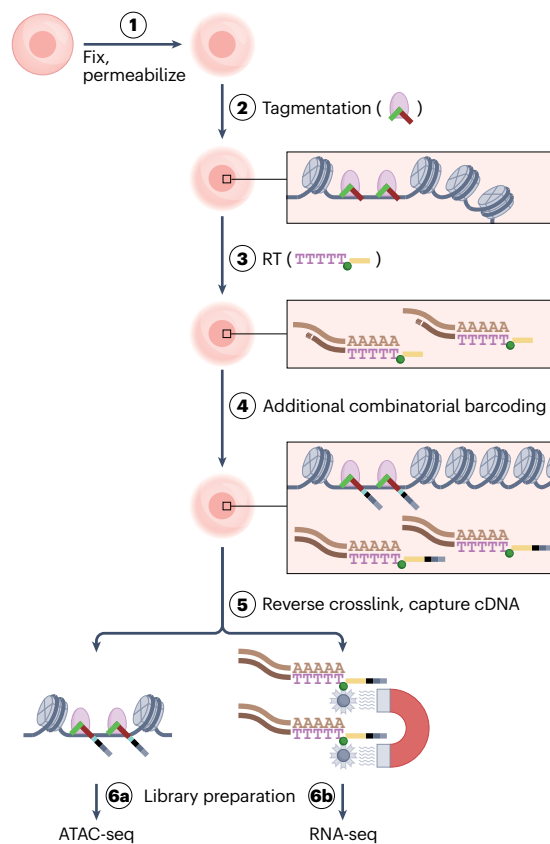
Most known layers of epigenetic information, including DNA methylation, chromatin accessibility, histone modifications, and binding of transcription factors (TFs) and chromatin remodelling complexes, can be recovered from single cells in parallel to the transcriptome. Such methods abound but range in sensitivity, specificity and ease of use (Fig. 1). The presence of epigenetic modifications can be read either directly from the DNA sequence, as is the case for DNA methylation, or indirectly by first encoding them in the sequence through DNA methylation and/or tagmentation. We discuss the latter approach first, as this strategy is most widely used.

Tagmentation-based methods come in a wide range of types. Most common are methods that jointly profile the transcriptome and

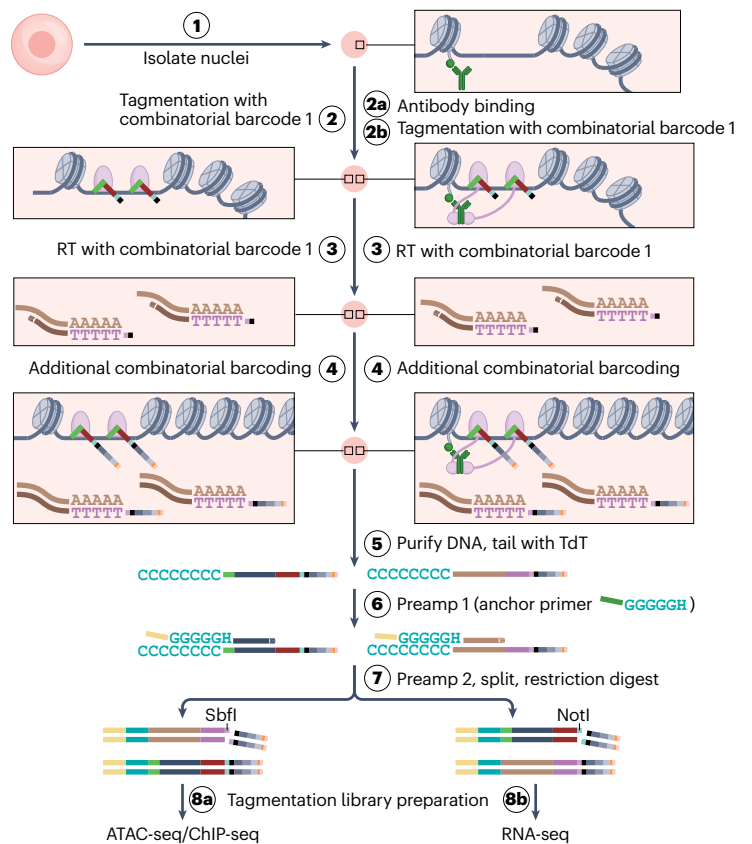
chromatin accessibility (Fig. 3). Here, accessible DNA is recovered as transposon-insertion-flanked regions using an assay for transposase-accessible chromatin (ATAC). Examples include scCAT-seq³⁵ and Smart3-ATAC³⁶, which are plate-based; sci-CAR-seq³⁷, SHARE-seq³⁸, SNARE-seq³⁹ and Paired-seq⁴⁰ (Fig. 3a,b), which rely on combinatorial indexing; SNARE-seq^{39,41}, ASTAR-seq⁴² and the commercially available 10x Genomics Multiome technology, which rely on microfluidics for cell barcoding (Fig. 3c); and ISSAAC-seq, which is amenable to both plate-based and microfluidics-based cell barcoding^{43,44}. scGET-seq represents an unusual type of such methods, profiling both accessible and inaccessible chromatin but not the transcriptome⁴⁵. Important features to be considered in tagmentation-based method selection include the ease of use of commercially available methods, the higher throughput of combinatorial indexing-based methods, the typical lower cost (but difficulty of establishment) of non-commercial, laboratory-developed methods, and the coverage, sensitivity and specificity obtained across cellular modalities. A systematic benchmark of these methods is unfortunately currently lacking. In most methods, transcriptome and accessible chromatin libraries are prepared in a common reaction and separated after indexing using magnetic beads, restriction enzymes or specific PCR primers (Fig. 3). A key advantage of the joint profiling of transcriptome and chromatin accessibility is that the link between gene expression and TF binding is more readily evaluable. In hair follicle cells, for example, SHARE-seq analyses of differentiation trajectories revealed TFs becoming expressed, with their activity being revealed in ATAC profiles later in pseudotime as binding sites became accessible, before expression of the associated target genes³⁸. As such, these multi-omic analyses in dynamic systems enable TF activity to be readily assigned to target genes. The ease of use of some of these methods has spearheaded their application in biomedicine and other domains.

However, the aforementioned methods quantify open chromatin without addressing the causes of accessibility changes, such as shifts in histone post-translational modifications (PTMs) or TF binding. To tackle this limitation, other methods have been developed in which

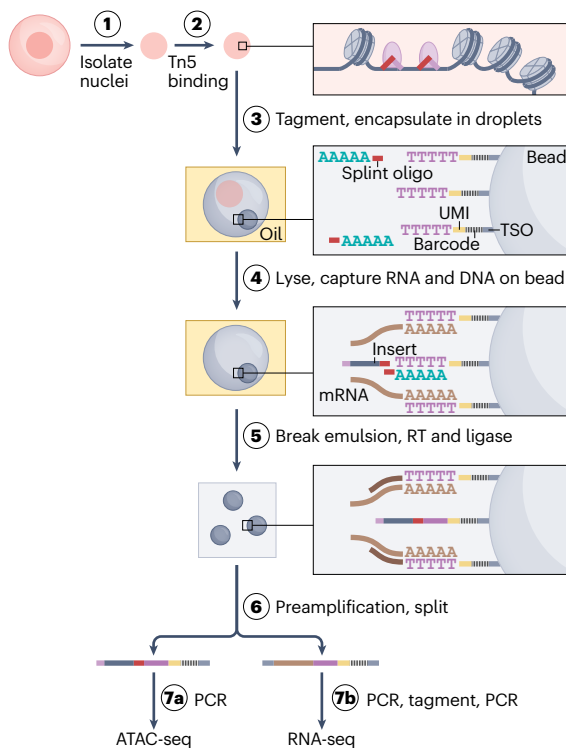
a SHARE-seq



b Paired-seq



c SNARE-seq



d TEA-seq

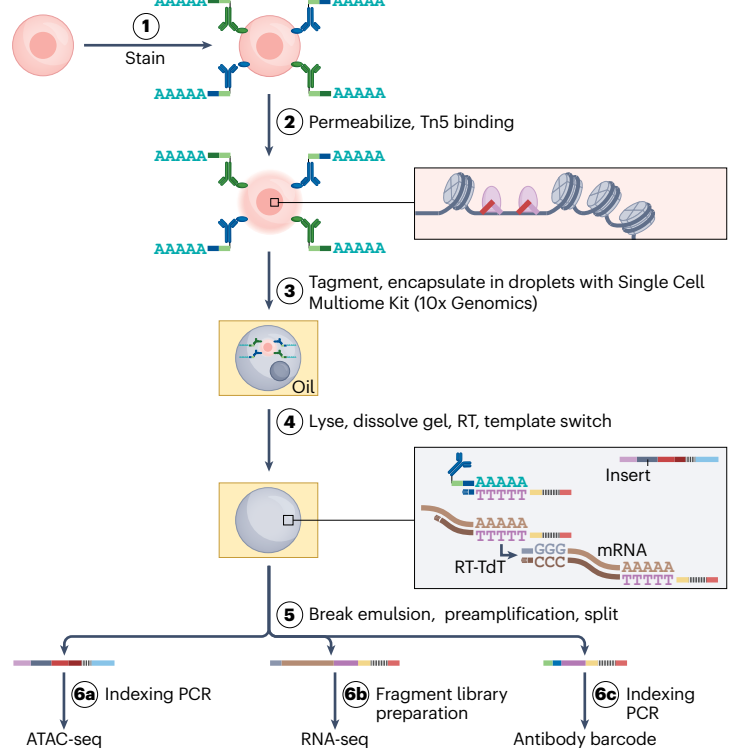


Fig. 3 | Selected tagmentation-based methods for single-cell multi-omic analyses. Summaries of experimental workflows highlighting how and in what order different modalities are probed and separated for analysis, while retaining single-cell information. Shown are SHARE-seq³⁸ (part a), Paired-seq²⁴⁰ (part b, left) and Paired-Tag⁴⁸ (part b, right), SNARE-seq³⁹ (part c) and TEA-seq⁸² (part d). In all methods shown, an assay for transposase-accessible chromatin (ATAC) precedes reverse transcription (RT) of polyadenylated RNA. Cell barcoding can occur through successive rounds of combinatorial indexing (parts a,b) or by compartmentalizing cells and barcoded oligonucleotides in microdroplets (parts c,d). DNA fragments originating from mRNA and DNA can be separated

by binding to paramagnetic beads (part a), differential restriction digestion (part b) or using specific PCR primers (parts c,d). Part b illustrates how similar workflows can either map accessible chromatin (left) and chromatin-associated proteins or their post-translational modifications (right). TEA-seq illustrates that barcoded oligonucleotides conjugated to antibodies can be detected using approaches similar to those developed for measuring gene expression. cDNA, complementary DNA; ChIP-seq, chromatin immunoprecipitation followed by sequencing; RNA-seq, RNA sequencing; TdT, terminal deoxynucleotidyl transferase; TSO, template-switching oligonucleotide; UMI, unique molecular identifier.

tagmentation is not randomly targeting accessible regions but directed towards specific histone PTMs or TFs. Conjugating the transposase to specific antibodies enables joint profiling of transcriptome and epitopes. Cell and modality barcoding can also occur through DNA–RNA separation, preamplification and split, seq-split or combinatorial indexing principles (Fig. 2), with transcriptome and DNA libraries being separated by restriction enzymes, PCR or beads. Examples of such methods include scPCOR-seq⁴⁶, coTECH⁴⁷, Paired-Tag (Fig. 3b)⁴⁸ and scSET-seq⁴⁹. scCUT&Tag2for1⁵⁰ and scMulti-CUT&Tag⁵¹ are distinct variations on these methods, involving two epitopes being targeted using different antibody-conjugated transposases. The distribution of both epitopes in a single cell can be learned from their different genomic distributions (for example, broad or narrow peaks) in scCUT&Tag2for1⁵⁰, or from epitope-specific barcoding enabled by transposases loaded with different oligonucleotides in scMulti-CUT&Tag⁵¹. A potent illustration of these methods was the profiling of mouse brain cells for both transcripts and different histone modifications using Paired-Tag, which identified distinct categories of genes each regulated by different epigenetic mechanisms⁴⁸.

Epigenetic information can also be read through DNA methylation profiling, relying either on 5-methylcytosine (5mC) or N⁶-methyladenine (6mA) (Fig. 4). 5mC is a prevalent endogenous epigenetic modification of DNA, found almost exclusively in a CpG context, that represses ectopic and heterochronic gene transcription initiation. In the most basic approach, endogenous 5mC DNA methylation is quantified in single cells by bisulfite sequencing. Key issues precluding more widespread adoption of single-cell 5mC analyses are the costs associated with library preparation and whole-genome sequencing, as well as the technical complexity. Indeed, bisulfite treatment of DNA causes its denaturation and fragmentation, and comes with a need for purification, leading to DNA losses. Library preparation can occur either through random-priming-based methods, such as post-bisulfite adapter tagging (PBAT), which is costly but yields genome-wide profiles covering 5–50% of the genome, or through reduced-representation bisulfite sequencing (RRBS), which is more cost-effective but only covers 1–3% of the genome. Recent evolutions in less destructive DNA methylome analytics and target capture may serve to develop more attractive workarounds^{52,53}. DNA methylation profiled on its own through single-cell DNA sequencing enables concomitant chromosomal copy number profiling. More often, DNA methylomes are profiled together with a cell's transcriptome, chromatin structure and/or chromatin accessibility.

Methods that produce transcriptome profiles alongside targeted DNA methylome profiles – as in scMT-seq, scTrio-seq and Smart-RRBS – or genome-wide DNA methylome profiles – as in scM&T-seq and scTrio-seq2 – have been established^{54–58}. These typically involve physical separation of DNA and RNA (Fig. 2a,b).

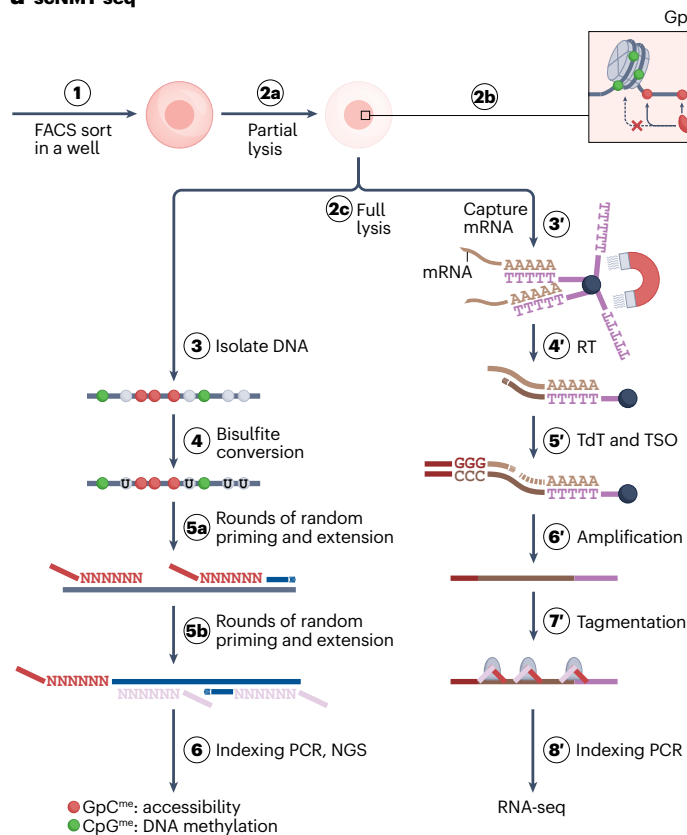
Endogenous CpG DNA methylation is often assessed in conjunction with chromatin accessibility. For this, a GpC methyltransferase is added to isolated nuclei where it can methylate open-chromatin-associated accessible DNA (Fig. 4a,b). Subsequent bisulfite sequencing can disclose the methylated GpC dinucleotides that mark these accessible regions, in addition to the endogenous methylation that in most cell types is nearly exclusive to CpG dinucleotides. GpC methylase-based methods show higher coverage per promoter than ATAC-based methods⁵⁹, albeit at substantially higher sequencing cost per cell, and make it easier to distinguish open from truly closed regions, which are not directly detected in the sparse scATAC-seq signals. A limitation relative to regular DNA methylome analyses is that at cytosines flanked on both sides by a guanine, methylation can be attributed to endogenous as well as exogenous processes. These are discarded from analyses. Relevant methods include scCOOL-seq⁶⁰, iscCOOL-seq⁶¹ and scNOME-seq⁶², which jointly profile accessibility and DNA methylation, and trimodal methods such as scNMT-seq⁵⁹, scNOMeRe-seq⁶³, scChARM-seq⁶⁴ and snmCAT-seq⁶⁵, which in addition uncover gene expression profiles from the same cell (Fig. 4a,b). All are plate-based assays and based on DNA–RNA separation, apart from snmCAT-seq, which is seq-split based (Fig. 2d; Fig. 4b).

Noteworthy alternatives that do not profile transcriptomes are snm3C-seq⁶⁶ and scMethyl-HiC⁶⁷, as they profile the DNA methylome in parallel to higher-order chromatin structure (Fig. 4c). Here, a 3C or HiC-like single-cell method to capture nuclear organization is combined with bisulfite conversion, to enable joint profiling of DNA methylation and chromatin structure^{66,67}. Methods that jointly also capture the transcriptome have yet to be described.

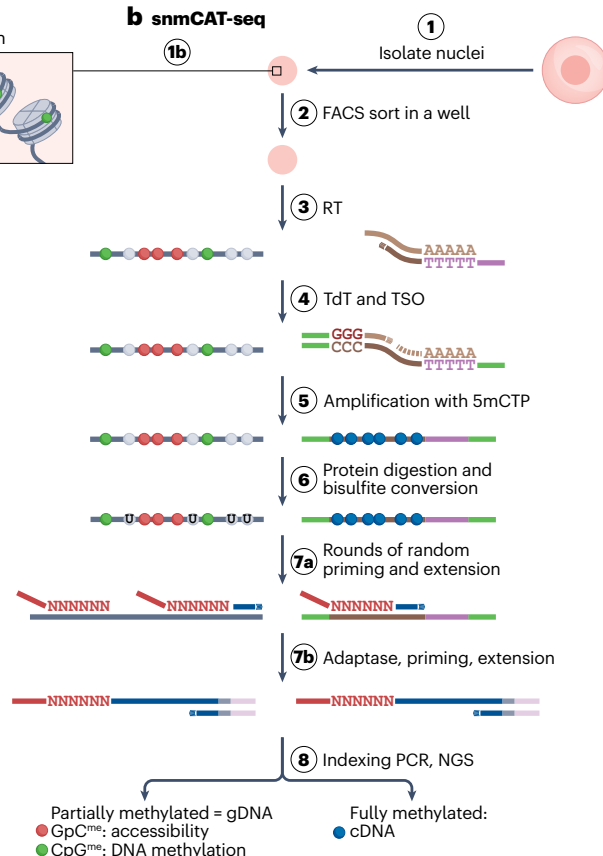
Despite being challenging techniques, the multi-omics nature of the resultant data enables a very in-depth analysis of cells, revealing hierarchies of changes to the epigenome and transcriptome during development, differentiation or pathogenesis. For example, stages of mouse gastrulation were profiled using scNMT-seq, which demonstrated that DNA methylation and chromatin accessibility patterns at ectodermal enhancers are pre-established in epiblast cells and stable during ectoderm differentiation, whereas mesodermal and endodermal enhancers are inactive in epiblast cells but actively remodelled following differentiation to mesoderm or endoderm⁶⁸. Accessibility and DNA methylation changes seemed to be tightly coordinated. These studies illustrate that multi-omics profiling of single cells for transcriptomic and epigenetic layers is feasible and provides fundamental insights unattainable using mono-omics methods.

Apart from methylated cytosines, methylation of adenines to 6mA can also be profiled in DNA (Fig. 4d). In contrast to 5mC, 6mA is an ultra-rare base in mammalian DNA⁶⁹ and can thus provide nearly unambiguous DNA marking when artificially introduced. Typically, a prokaryotic DNA adenine methyltransferase (Dam) is used to methylate adenines in GATC context for DNA-sequence encoding of

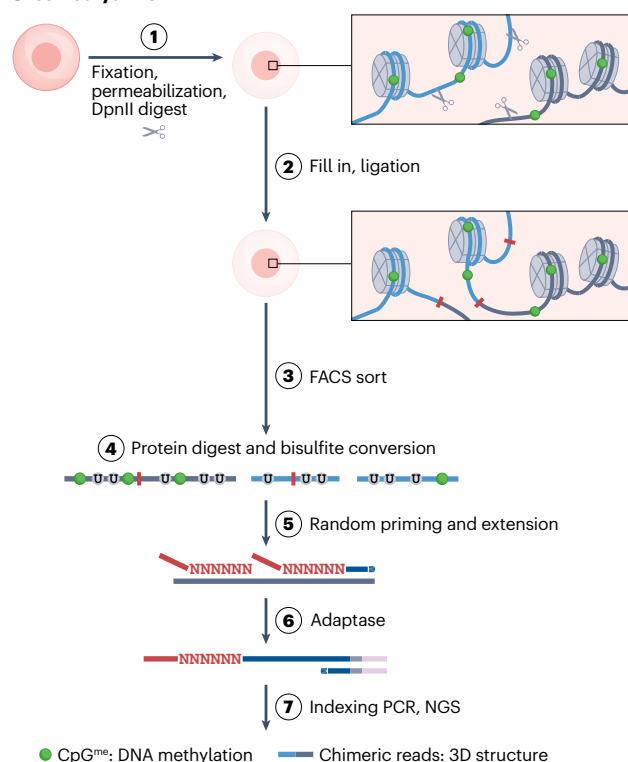
a scNMT-seq



b snmCAT-seq



c scMethyl-HiC



d EpiDamID with scDam&T-seq

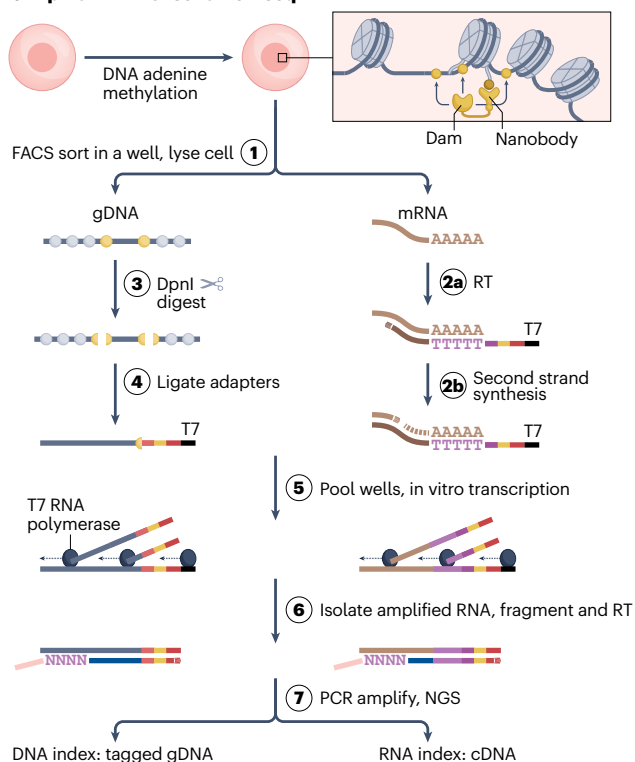


Fig. 4 | Selected DNA methylation-based methods for single-cell multi-omic analyses. Summaries of experimental workflows highlighting how and in what order different modalities are probed and separated for analysis, while retaining single-cell information. Shown are scNMT-seq⁵⁹ (part **a**), snmCAT-seq⁶⁵ (part **b**), scMethyl-HiC⁶⁷ (part **c**) and EpiDamID with scDam&T-seq⁷³ (part **d**). In all methods shown, several types of epigenetic information can be discerned from a single sequencing library: DNA methylation and accessibility (part **a**), DNA methylation, accessibility and gene expression (part **b**), DNA methylation and chromatin conformation (part **c**), and histone modifications and transcription (part **d**). These layers of information can be discriminated by analysing DNA methylation patterns (parts **a,b**), DNA methylation and read-pair mapping (part **c**) or read-associated barcode tags (part **d**). Only scNMT-seq (part **a**)

epigenetic information. Dam can either be expressed to mark accessible chromatin^{70–72} or be tethered to endogenously expressed proteins using insertion mutagenesis to mark their binding sites in live cells. Two DNA–RNA separation-based methods applying adenine methylation to 6mA have been described: scDam&T-seq, with Dam tethered to chromatin-associated proteins to mark their nuclear location (DamID)⁷⁰, and scDam&T-seq with EpiDamID, in which Dam is tethered to protein domains or nanobodies that recognize PTMs⁷³ (Fig. 4d). As 6mA cannot be detected directly through short-read sequencing, quantification is relative, relying on Dam activity and a 6mA-specific restriction digest. In contrast to the other chromatin profiling methods described above, the resulting profiles typically reflect the aggregate of residence times of the proteins marked, leading to better signal-to-noise ratios but poorer temporal resolution. The reliance on endogenous tethering limits the scope of these methods to systems amenable to transgenesis. Interestingly, it extends their scope beyond only those proteins for which a specific antibody is available, and at least scDam&T-seq enables a method of analysing chromatin occupancy that is orthogonal to antibody-based methods. Additionally, these methods may suffer less from the bias towards accessible chromatin that characterizes transposase-based tagging approaches. Interestingly, these methods share several features with Fibre-seq-based approaches⁷⁴, in which chromatin patterns are read through 6mA marking using single-molecule long-read sequencing. Combining both methods may thus facilitate high-throughput, haplotype-resolved and cell-type-specific characterization of chromatin structures in bulk.

Single-cell omics plus low-plex profiling of another analyte

Aside from profiling multiple comprehensive omics layers from the same cell, substantial progress was recently made in profiling single cells comprehensively for a single analyte and in a less comprehensive (low-plex) manner for another analyte. Most common among these methods are those quantifying the cell's transcriptome and/or accessible genome, alongside a limited set of cell surface or intranuclear proteins. These methods typically rely on antibodies tagged with a specific barcoded oligonucleotide, which can be captured and amplified alongside the transcriptome. Using mixes of dozens to hundreds of such barcoded antibodies that each recognize specific epitopes, subsequent barcode counting thus enables quantification of multiple proteins of interest in single cells alongside true omics analysis. Such methods enable the profiling of protein abundances in addition to gene expression or other modalities, but can also capture other information about proteins such as protein stability, PTMs and protein isoform expression. Omics layers shown to be amenable to this include the transcriptome (REAP-seq⁷⁵, CITE-seq⁷⁶, inCITE-seq⁷⁷, SPARC⁷⁸, ECCITE-seq⁷⁹ and RAID-seq⁸⁰), open chromatin (ASAP-seq⁸¹ and ICICLE-seq⁸²) or both (DOGMA-seq⁸¹

involves physical separation of modalities to be probed (mRNA from intact cells). Each of these methods is plate-based and therefore restricted in its throughput. DNA methylation patterns can be read using bisulfite conversion of unmethylated cytosines to uracil (U) (parts **a–c**) or methylation-sensitive restriction digestion (part **d**), and reflect endogenous methylation alone (part **c**), endogenous methylation as well as exogenously added methylation reflecting chromatin accessibility (parts **a,b**), or endogenous adenine methylation added after genetic transformation (part **d**). cDNA, complementary DNA; Dam, DNA adenine methyltransferase; FACS, fluorescence-activated cell sorting; gDNA, genomic DNA; 5mCTP, 5-methyl-deoxycytidine triphosphate; NGS, next-generation sequencing; RT, reverse transcription; T7, T7 promoter; TdT, terminal deoxynucleotidyl transferase; TSO, template-switching oligonucleotide.

and TEA-seq⁸²), as well as chromatin modifications (scCUT&Tag-pro⁸³) (Fig. 3d). Epitopes available for profiling are mostly limited to the cell surface, although methods for intracellular (SPARC⁷⁸ and RAID-seq⁸⁰) and intranuclear (inCITE-seq⁷⁷ and NEAT-seq⁸⁴) epitopes have also been developed. Note that any plate-based method can theoretically also leverage antibody marking and cytometry to quantify a limited set of proteins per cell. A notable alternative approach (PHAGE-ATAC) was described recently in which barcoded phages that display a nanobody serve to bind cell surface epitopes⁸⁵. The barcoded phage genomes can be quantified in conjunction with genome-wide chromatin accessibility profiles. Each of these methods is currently limited by the availability of specific antibodies or nanobodies, and although mass spectrometry-based methods for proteome-wide analysis of single cells have been developed⁸⁶, such analyses alongside other omics layers are currently lacking. A related method recently described is single-cell transcriptome and translational sequencing (T&T-seq), in which cells are sorted into plates and lysates are distributed for total RNA-seq and for affinity purification of actively translating ribosomes. T&T-seq thus enables joint profiling of all transcripts and those transcripts that are being translated into proteins⁸⁷.

Finally, other molecular features have also been shown to accommodate barcoding, with guide RNAs as a key example. In cells expressing CRISPR-based gene editing, activation or inactivation systems, feature barcoding methods enable high-throughput profiling of the transcriptome or accessible chromatin in pools of single cells subjected to high-throughput genetic perturbation screens. Examples include CROP-seq, in which the guide RNA sequence is directly determined alongside the cell's transcriptome⁸⁸, and CRISP-seq⁸⁹ and Perturb-seq^{90,91}, in which each guide RNA has a unique barcode that is sequenced alongside the transcriptome. The latter two methods have lower specificity, as recombination can blur the link between barcode and guide RNA, and more recent implementations of Perturb-seq therefore apply direct guide-RNA sequencing. Methods to profile the impact of such perturbations on chromatin accessibility include CRISPR-sciATAC⁹², Perturb-ATAC⁹¹ and Spear-ATAC⁹³. Recently, this approach was applied at scale when millions of single-cell transcriptomes were analysed, with each expressed gene inactivated in a subset of these cells. This strategy revealed the effect of inactivation of each expressed gene on the expression of all other genes, and thus represents a valuable resource for the *in silico* modelling of genetic perturbations⁹⁴. Notably, a similar approach can be deployed to characterize gene interactions⁹⁵. Here, cells are transfected with, on average, two different guide RNAs from a pool, generating pools of double-knockout cells. The combined impact of both perturbations can then be compared with the impact of single perturbations, providing direct quantification of a matrix of genetic interactions.

Box 1

Principles of spatial mono-omics methods for spatial multi-omics developments

Methods for spatial mono-omics have progressed tremendously in the last decade with excellent reviews existing for spatial transcriptomics^{165,171–173}, (epi)genomics^{166,174}, proteomics^{86,167} and metabolomics^{168,175}. Here, we briefly highlight the basic principles of the approaches that currently form the basis for most ongoing spatial multi-omics developments.

Array-based spatial transcriptomics

These approaches allow for transcriptome-wide profiling through capturing of polyadenylated (poly(A)) RNA transcripts released from fixed and permeabilized tissue sections, making use of thousands of arrayed and spatially barcoded oligo-dT spots (for example, Spatial Transcriptomics^{170,176} and 10x Genomics Visium¹¹⁵), uniquely DNA-barcoded beads (for example, Slide-seq¹⁷⁷ and HDST¹⁷⁸) or even barcoded DNA nanoballs (for example, Stereo-seq¹⁷⁹) on a slide surface. Following reverse transcription into barcoded cDNA and library preparation, gene expression is spatially profiled by addressing the barcoded next-generation sequencing (NGS) reads to specific locations within the imaged tissue section.

Microfluidic deterministic barcoding strategies

These methods (including DBiT-seq¹⁰³, spatial-ATAC-seq¹⁰¹ and spatial-Cut&Tag¹⁰²) allow high-resolution NGS-based spatial (multi-)omics profiling. By serially attaching two polydimethylsiloxane (PDMS) microfluidics chips with equidistant microchannels perpendicular to each other on a pre-fixed tissue section, two sets of channel-specific DNA barcodes (A1-N and B1-N) can be delivered to the tissue surface for barcoding of captured poly(A) transcripts, proteins or epigenomic information. This yields a 2D map of pixels on the tissue with unique barcode combinations (AB) at the channel crossflows to spatially allocate barcoded NGS reads for the analytes of interest.

DNA antibody tags

This approach allows protein targeting and spatial mapping via NGS by using polyadenylated antibody-derived tag-conjugated antibodies, derived from single-cell CITE-seq⁷⁶. These antibodies linked to DNA sequences with an antibody-specific barcode and a poly(A) tail are compatible with the capture-based array or microfluidic deterministic barcoding approaches above, for spatial (co-)profiling of proteins by NGS. Alternatively, next to the imaging-based methods that rely on cyclic fluorescent antibody stains for the identification of up ~60 protein targets in tissue sections (such as seqIF¹⁸⁰, CycIF¹⁸¹ and MILAN¹⁸²), antibodies labelled with DNA barcodes can be identified

through cyclic hybridization and imaging of fluorescent readout sequences or with fluorescent detection during oligosequence amplification (for example, CODEX¹⁸³ and immuno-SABER¹⁸⁴).

Multiplex smFISH

Multiplex single-molecule fluorescent in situ hybridization (smFISH) methods allow imaging of thousands of gene transcripts and genomic loci in single cells with high accuracy and subcellular resolution, to enable single-molecule detection of RNAs in their native tissue context and characterize chromosomal structure and organization. With the most advanced methods (such as MERFISH^{107,111} and seqFISH+^{112,185}), predefined optical barcoding schemes are assigned and imprinted onto oligonucleotide targets using a library of encoding probes. Each encoding probe contains a region targeting the sequence of interest and variable distinct readout sequences in multiple copies. After encoding probe binding, detection of the imprinted barcodes is achieved through multiple rounds of fluorescent readout probe hybridization, high-resolution imaging and signal quenching.

In situ sequencing

In situ sequencing (ISS) methods allow high-throughput spatial mapping of transcripts and genomic loci in single cells. Both targeted and untargeted approaches (for example, ISS¹⁸⁶, ExSeq¹²⁵ and FISSEQ^{187,188}) exist for transcript identification, in which highly specific padlock probes are hybridized to in situ-synthesized cDNA sequences, followed by probe ligation and rolling circle amplification (RCA) of either the barcoded probe sequences or short sequences of the cDNAs, to generate micrometre-sized RCA products within cells that are decoded using in situ imaging-based sequencing-by-ligation. OligoFISSEQ¹¹³ methods also allow genome-wide targeting of genomic loci.

Mass spectrometry imaging

Mass spectrometry imaging (MSI) is an alternative to the antibody-based approaches for spatially characterizing not only proteins or peptides but also other small biomolecules, such as lipids, metabolites or sugars, in tissue sections. Samples are systematically scanned by light or particle beams ionizing the surface biomolecules and making them accessible to time-of-flight (TOF) mass spectrometry for identification. Matrix-assisted laser desorption/ionization (MALDI) MSI methods enable spatial metabolome characterization in tissue sections at (near) single-cell resolution¹⁸⁹.

Spatial multi-omics methods

Methodologies for spatial multi-omics are developing rapidly to allow the study of different molecular analytes at up to subcellular resolution within their native tissue context (Fig. 1). Spatial multi-omics technologies were listed by *Nature* as one of the seven technologies to watch in 2022⁹⁶, with the basis for their development and ongoing innovations

being a range of established spatial mono-omics methods (Box 1). Spatial multi-ome characterization of a sample, usually a fixed fresh-frozen or formalin-fixed paraffin-embedded (FFPE) tissue section, is often achieved by combining these spatial mono-omics methods. They can be applied separately on adjacent tissue sections, serially on the same tissue section if the quality of the different analytes can be maintained,

or in parallel on the same tissue section if joint targeting and reading out of the different analytes is possible. The number of target analytes that can be analysed simultaneously at the different molecular levels varies between methodologies (Fig. 1). Often spatial omics measurements are also supplemented with histological stains, such as H&E (haematoxylin and eosin) staining, of the same or adjacent tissue sections, allowing integration with additional morphological annotations.

Spatial multi-omics via adjacent-section strategies

The application of spatial assays for different mono-omics layers of interest (Box 1) on adjacent or serial sections from the same tissue sample enables these techniques to be assayed in their most optimal setting and their data to be integrated computationally (Fig. 5a). This requires sample collection and preparation procedures compatible with all assays of interest, well-thought-out experimental design and good assay coordination. This strategy is mostly required when multi-omics readouts of interest are not possible yet or suboptimal on the same section. Mass spectrometry imaging (MSI) methods⁹⁷, for example, can be used for spatial metabolome characterization of small biomolecules such as lipids (Box 1), but cannot easily be combined with other spatial genome, transcriptome or proteome readouts on the same section owing to specific sample preparations and limitations (for example, the need for matrix deposition in MSI⁹⁷, OCT compound embedding complicating mass spectrometry⁹⁸ and inherent breakdown of lipids during potential upfront assays). Conversely, MSI is compatible with H&E staining on the same section, which can also be leveraged for computational integration with other spatial assays. Additional serial sections can also be dissociated into single cells or nuclei for generating matched single-cell sequencing data that can be used for optimal deconvolution of the spatial data and additional data integration (Fig. 5a). As such, a combination of different single-cell sequencing and spatial transcriptomics and proteomics approaches was applied and integrated to assemble an atlas of healthy and obese murine and human livers and to map hepatic macrophage niches⁹⁹. Disadvantages of this approach are that not all assays are compatible with all sample types (for example, fresh-frozen versus FFPE), that it suffers from sample heterogeneity, as even adjacent sections will differ slightly as to structure and cellular composition, and that different spatial assays may have different resolutions. Although computational tools exist that can compensate for these challenges (see below), the concordance between sections is never unambiguous. Hence, innovative methods enabling spatial multi-omics measurements on the same tissue sections have started to emerge.

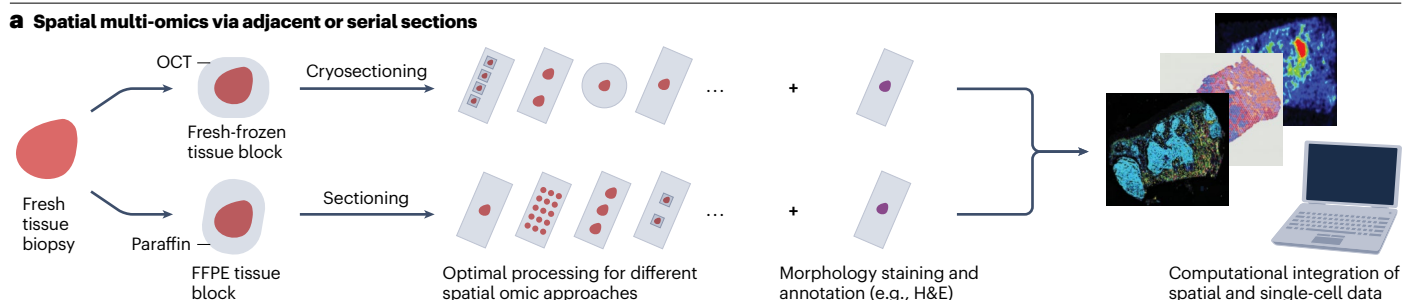
Spatial (epi)genomics-plus-transcriptomics

The possibility for simultaneous unbiased profiling of chromatin accessibility or specific histone modifications and gene expression on the same tissue cryosections has been described for spatial ATAC&RNA-seq and spatial CUT&Tag-RNA-seq, respectively¹⁰⁰. These methods are based on combining microfluidic deterministic barcoding in tissue (DBiT) strategies (Box 1) for spatial-ATAC-seq¹⁰¹ or spatial-CUT&Tag¹⁰² with DBiT-seq poly(A) transcript profiling¹⁰³ (Fig. 5b). For capturing chromatin accessibility, accessible gDNA is first tagged in situ with a universal ligation linker. For capturing specific histone modifications, the tissue is first incubated with primary antibodies against the epigenetic marks, followed by a secondary antibody that allows protein A-transposome tethering for fragmentation of the DNA at these specific locations. Combined mapping of the transcriptome is achieved by hybridizing a biotinylated oligo-dT that also contains

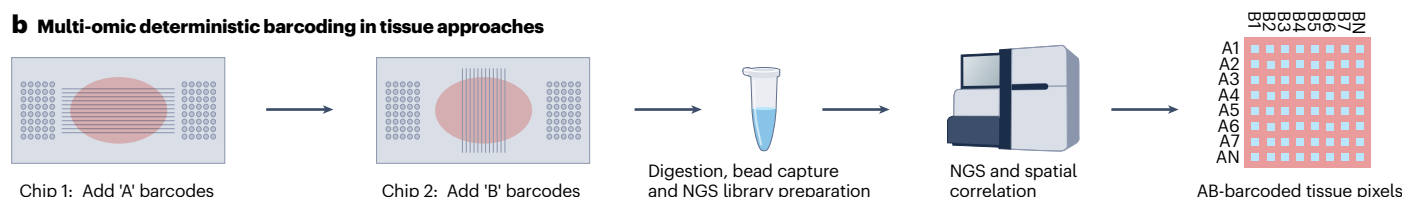
a universal ligation linker and primes RT in situ¹⁰⁰ (Fig. 5b). DBiT-seq barcoding with serial attachment to the tissue of two microfluidics chips with equidistant channels perpendicular to each other is then used to administer two sets of channel-specific barcodes (Box 1), with the first set 'A' ligating with the universal ligation linkers present on the tagged DNA and poly(A) mRNA-derived cDNA, and the second set 'B' ligating to the set 'A' barcodes, resulting in an in situ 2D grid with uniquely barcoded AB crossroads (20–25-μm pixels)¹⁰⁰. The spatially barcoded gDNA and cDNA fragments are collected by reverse crosslinking, cDNA is enriched with streptavidin-coated magnetic beads, gDNA is retained in the supernatant, and NGS libraries are constructed separately for sequencing (Fig. 5b). Based on the spatial barcodes, sequencing reads are combined with microscopy images of the tissue section, allowing the multi-omics sequence information to be mapped spatially¹⁰⁰. Limitations of these assays are the near-single-cell resolution (20–25-μm pixel size), the small analysable area (2,500–10,000 pixels), the uncharacterized spaces in between adjacent pixels (depending on channel distances) and the expertise that is required in fabricating and handling microfluidics chips for implementation. Nevertheless, these methods were successfully applied to developing mouse and adult human brains, revealing how epigenetic states or modifications regulate cell type, states and dynamics¹⁰⁰.

Alternatively, microscopy-based methods can enable spatial profiling of genome or epigenome information together with gene expression by directly imaging DNA loci, chromosomal and nuclear structures, and transcripts within single cells at up to subcellular resolution. Several methods showing imaging-based multi-omic measurements for a limited number of combined analytes exist^{104–106}, but we focus on those that can characterize at least one analyte layer more comprehensively. Multiplex single-molecule fluorescent in situ hybridization (smFISH) methods, such as MERFISH^{107,108} and seqFISH^{109,110} approaches (Box 1), originally designed for targeted high-sensitivity spatial profiling of thousands of transcripts at subcellular resolution, were adapted to allow genome-scale chromatin tracing as in DNA-MERFISH¹¹¹ and DNA-seqFISH+¹¹². Combined RNA, chromatin and nuclear body imaging in the same fibroblast cells was enabled by sequentially staining for >1,100 nascent transcripts using RNA-MERFISH, followed by DNA-MERFISH for >1,000 genomic loci, and finally fluorescent antibody readouts for cell-cycle-state determination and landmark nuclear structures, including nuclear speckles and nucleoli¹¹¹ (Fig. 5c). This approach allows for the characterization of chromatin domains, compartments and trans-chromosomal interactions and their relationship to transcription in single cells¹¹¹. Similarly, DNA-seqFISH+ imaging of up to 3,660 chromosomal loci in mouse embryonic stem cells was shown to be compatible with RNA-seqFISH of 70 mRNAs and intron-seqFISH of 1,000 genes at their nascent transcription active sites, as well as sequential immunofluorescence targeting of 17 nuclear structures, including the nuclear lamina, nucleolus and histone modification marks, using primary antibodies conjugated with DNA-oligonucleotides detectable by fluorescently labelled readout probes¹¹² (Fig. 5c). In addition to identifying heterogeneity in chromosome structure, this study found that many active gene loci reside at the surface of nuclear bodies and the presence of persistent global chromatin states¹¹². Also, OligoFISSEQ methods allow rapid in situ sequencing-based visualization (Box 1) of multiple genomic loci in single cells, with the potential for genome-wide application and being compatible with immunofluorescence and other FISSEQ-based methods for protein and RNA characterization, respectively¹¹³. Although these microscopy-based methods are important for studying chromosomal and nuclear structures and organization linked to gene

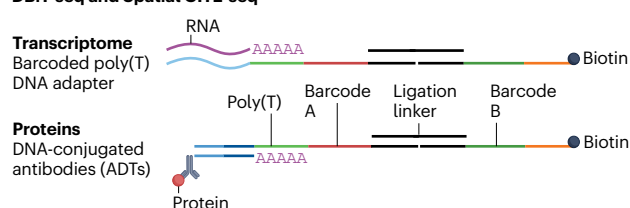
a Spatial multi-omics via adjacent or serial sections



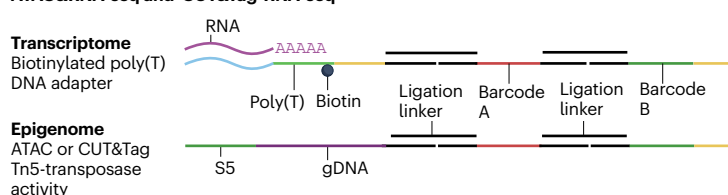
b Multi-omic deterministic barcoding in tissue approaches



DBiT-seq and Spatial CITE-seq



ATAC&RNA-seq and CUT&Tag-RNA-seq



c Multi-omic single-molecule fluorescent in situ hybridization approaches

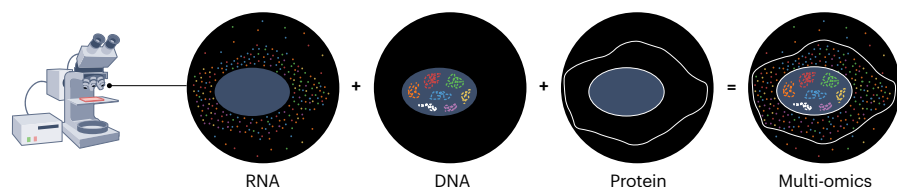
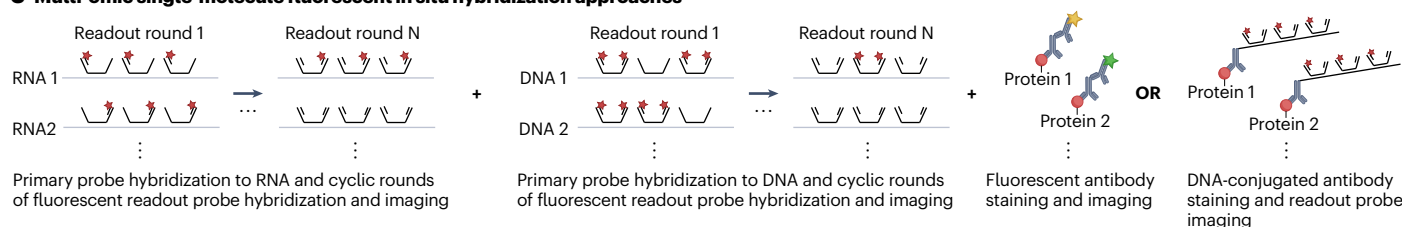
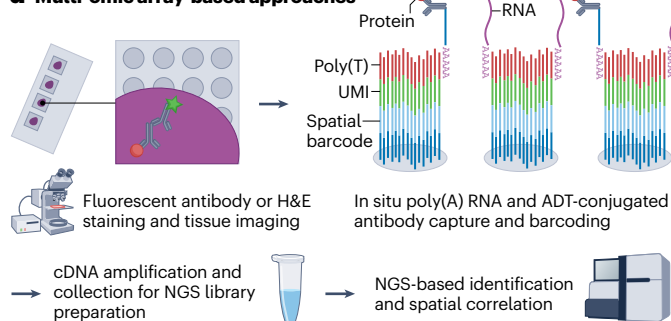


Image registration and decoding of optical barcodes

Readout round	1	2	3	4	5	...	N ₂	N ₁	N
Target analyte 1	1	0	0	0	1	...	0	0	1
Target analyte 2	1	0	0	0	0	...	0	0	0
⋮									

d Multi-omic array-based approaches



e Multi-omic Digital Spatial Profiling approach

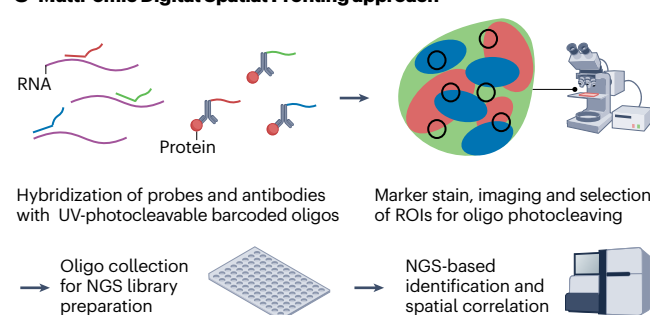


Fig. 5 | Methods for spatial multi-omics. Spatial multi-ome profiling of tissue samples can be achieved by applying spatial mono-omics assays separately on adjacent or serial tissue sections (part **a**) or in a combined way on the same tissue section (parts **b–e**). **a**, Serial fresh-frozen or formalin-fixed paraffin-embedded (FFPE) tissue sections can be analysed using different spatial mono-omic assays, potentially also combining with morphological stainings and annotations on the same or adjacent sections, followed by computational data integration. **b**, Microfluidic deterministic barcoding strategies in tissue allow next-generation sequencing (NGS)-based spatial multi-omics profiling of transcriptome-plus-proteins, as in DBiT-seq¹⁰³ and Spatial-CITE-seq¹²², and epigenome-plus-transcriptome, as in ATAC&RNA-seq and CUT&Tag-RNaseq¹⁰⁰. Using dual microfluidic chip-based spatial barcoding of poly(A) RNAs together with proteins or epigenome information at the crossroads of chip channels, a spatially barcoded 2D pixel map of the tissue is created. **c**, Advanced fluorescence in situ hybridization (FISH)-based methods, including MERFISH^{107,108,111} and seqFISH+^{109,110,112}, allow microscopy-based identification of thousands of transcripts together with genomic loci in single cells, in addition to being compatible with limited protein readouts using fluorescent or DNA-conjugated

antibody readout strategies. These high-resolution imaging methods leverage predefined optical barcoding schemes and complex encoding and readout probe designs. **d**, Array-based assays, including Spatial Transcriptomics¹⁷⁰ (ST) and 10x Genomics Visium¹¹⁵, make use of slides with arrayed oligo-dT spots for capturing and spatial barcoding of poly(A) RNAs followed by NGS profiling. This can be combined with upfront haematoxylin and eosin (H&E) staining or limited protein antibody staining and tissue imaging for spatial mapping. In SM-Omics¹¹⁷ and SPOTS¹¹⁶, these technologies have also been shown to be compatible with antibody-derived tag (ADT)-conjugated antibody-based co-profiling of a larger number of proteins. **e**, NanoString GeoMx digital spatial profiling (DSP)^{118,119,121} allows quantification of RNAs and proteins in specific regions of interest (ROIs) by counting uniquely barcoded oligonucleotides that are covalently linked through a UV-photocleavable linker with probes or antibodies. Tissue marker staining, imaging, ROI selection and illumination by directed UV light causes disintegration of the photocleavable linkers that are collected and profiled by NGS, followed by spatial mapping to the ROIs. cDNA, complementary DNA; gDNA, genomic DNA; OCT, optimal cutting temperature compound; UMI, unique molecular identifier.

expression regulation at a single-cell level, they require expertise in complex optical barcoding schemes and high-resolution imaging modalities, are challenging to apply in complex tissue samples, are costly and usually can only characterize a limited area.

Alternatively, by using laser capture microdissection (LCM)-based isolation of specific (single) cells from tissue sections, single-cell sequencing-based genome-plus-transcriptome or epigenome-plus-transcriptome profiling methods as described above can be applied at spatial resolution, as was done for analysing tumour development, metastasis and prognosis in patients with triple-negative breast cancer¹¹⁴.

Spatial transcriptomics-plus-proteomics

Methodologies allowing parallel spatial interrogation of both the transcriptome and proteome are currently still limited, are often based on serial characterization of both modalities, mostly allow co-characterization of only a limited number of proteins and often lack single-cell resolution. For example, the commercial array-based 10x Genomics Visium technology for poly(A) RNA capture and spatial barcoding at 55-µm resolution followed by NGS identification (Box 1) currently supports immunofluorescence protein detection of one or two targets on the same fresh-frozen or FFPE tissue section, although at the cost of the otherwise applied H&E staining used for spatial mapping¹¹⁵ (Fig. 5d). However, Spatial Proteome and Transcriptome Sequencing (SPOTS) demonstrates that Visium is also compatible with co-profiling of a larger number of proteins using polyadenylated antibody-derived tag (ADT)-conjugated antibodies (Box 1), as was shown for 21 proteins in mouse spleen and breast cancer tissue cryosections¹¹⁶. Similarly, Spatial Multi-Omics (SM-Omics) shows Spatial Transcriptomics (the forerunner of Visium with 100-µm resolution) to be compatible with DNA-barcoded antibody and/or immunofluorescence co-mapping of six proteins, with the added concept of full automation using liquid-handling robotics achieving increased throughput and performance¹¹⁷.

Alternatively, NanoString GeoMx Digital Spatial Profiling (DSP) allows quantification of the abundance of RNAs and/or proteins by counting unique indexing oligonucleotides, which are covalently linked via a UV-photocleavable linker with probes or antibodies that target transcripts or proteins of interest, respectively^{118,119} (Fig. 5e). Whole-transcriptome DSP assays or DSP assays for >100 protein targets are possible, although coupled to image-based selection of specific regions

of interest following fluorescent antibody and/or RNAscope¹²⁰ marker stainings. Selected regions of interest are illuminated by directed UV light, which cleaves the photocleavable linkers and releases oligonucleotide indices, enabling their collection and identification using NGS. Achieving single-cell resolution is challenging with this technology, but the combination of both transcriptome and protein readouts on the same sample was shown in FFPE cell pellet array sections stained with the GeoMx Human Whole Transcriptome Atlas (WTA) probe set and a 59 GeoMx antibody panel¹²¹.

Chip-based DBiT-seq approaches (Box 1) also allow co-mapping of poly(A) mRNAs with proteins in tissue cryosections, as shown in the original DBiT-seq method with ~10-µm pixel resolution and tens of proteins co-profiled¹⁰³, but improved upon in spatial-CITE-seq (spatial co-indexing of transcriptomes and epitopes), in which whole-transcriptome and co-mapping of ~200–300 proteins was achieved at 20-µm pixel resolution¹²² (Fig. 5b). Both these methods employ cocktails of ADT-conjugated antibodies (Box 1) each bearing a poly(A) oligonucleotide with an antibody-specific barcode. The channel-specific set 'A' barcodes, delivered by the first chip with equidistant channels placed on the tissue, consist of barcoded poly(T) DNA adapters also containing a universal ligation linker that hybridizes to both poly(A) ADT and mRNA molecules, and are then incorporated during *in situ* RT. The second chip introduces perpendicularly a set of channel-specific set 'B' barcodes that are ligated to the 'A' barcodes and also contain a PCR handle functionalized with biotin. The specific 'AB' barcoded cDNA and ADTs at the channel crossroads (pixels) are collected, purified with streptavidin-coated magnetic beads and prepared for NGS, after which reads for both mRNAs and proteins can be superimposed on an image of the section¹⁰³ (Fig. 5b). Spatial-CITE-seq was applied for improved tissue mapping to several mouse tissue types, as well as human tonsillar and skin biopsy tissue at the COVID-19 mRNA-vaccine injection site¹²². Although limited by the same aspects as previously mentioned for DBiT-based methods, including lack of single-cell resolution, to date spatial-CITE-seq is the method that allows for the highest number of proteins to be simultaneously spatially profiled together with the poly(A) transcriptome, with potential for further expansion.

As to microscopy-based approaches, several of the established seqFISH-based, MERFISH-based and *in situ* sequencing (ISS)-based

methods for spatial transcript profiling of hundreds to thousands of targeted genes (Box 1) are also (sequentially) compatible with immunofluorescence or DNA-conjugated antibody protein readouts in the same sample (Fig. 1). For the latter, oligonucleotide sequences compatible with the specific fluorescent readout approaches of the transcriptome profiling methods are mostly used (Fig. 5c)^{107,111,113,123–125}. Staining and localization of cell boundaries or other cellular, nuclear or subnuclear markers in these ways can enable more accurate cell segmentation, transcript allocation or resolution of nuclear organization. Additionally, combined spatial transcriptome and protein readouts on the same tissue section are crucial to correlate transcript with protein expression, localization and interactions, to help unravel the cellular mechanisms that govern specific cell types and states. Several platforms that recently became commercially available for automated imaging-based spatial transcriptome profiling of tissue sections at single-cell resolution, including the NanoString CosMx (smFISH-based)¹²⁶, Vizgen MERSCOPE (MERFISH-based)¹²⁷ and 10x Genomics Xenium (ISS-based)¹²⁸ platforms, will allow for imaging-based co-profiling of a few to tens of proteins, making these technologies more accessible.

Data integration

A wide variety of algorithms have been described for the tailored data analysis of individual analytes in single cells. Here, we focus on computational advances that allow the integration of measurements across several modalities, as these algorithms leverage the multi-omic nature of these technologies to enhance our understanding of complex cellular states. Each modality covers different aspects of cellular identity and has its strengths and weaknesses. A principal goal of multi-omics data integration is to achieve robust and sensitive cell type or cell state identification¹²⁹. This integrated multi-omics view of cellular identity can improve our understanding of differentiation trajectories, their underlying gene regulatory networks, cell–cell interactions, micro-environmental spatial organization, cellular lineages and clonal dynamics. Ultimately, a holistic view of cellular identity disentangles causal relationships between the different molecular layers that give rise to the observed cellular phenotypes. However, to achieve meaningful integration of high-dimensional data modalities, computational and statistical models need to be developed that consider the technical and biological intricacies of these technologies.

Recently, Argelaguet et al.¹³⁰ defined three categories of data integration strategies depending on the anchor used to link the different data modalities. Horizontal integration strategies use common data features measured across different datasets to integrate independently assayed groups of cells, such as when assaying different batches with the same technology or when integrating across different technologies measuring the same molecular analyte. Inversely, vertical integration strategies use the cell as the anchoring unit to integrate non-overlapping data features, such as when measuring multiple omics layers of the same cells in parallel (Fig. 6a). The hardest integration problem emerges when neither cells nor common data features can be used as anchors. In this case, diagonal integration strategies are used to map separate groups of cells profiled by different molecular assays (Fig. 6b). Many computational methods have already been developed to tackle horizontal integration problems, as cell atlas efforts require the integration of many batches of single-cell data. For a comparison of these approaches, we refer to recent reviews and benchmark studies^{130–133}. Here, we focus primarily on vertical integration approaches as these are applicable to most multi-omics technologies in which parallel measurements are made of individual cells.

Linking molecular layers

When paired data modalities can be unambiguously assigned to individual cells, several integration strategies can be applied depending on the end goal of the analysis. For finding significant cross-modal associations between specific features (coined local integration methods by Argelaguet et al.¹³⁰), classic regression-based statistical models can be fitted in a supervised way¹³⁴. These models can be expanded to handle common confounders (such as population stratification biases, sample-specific effects and sequence context-dependent effects), which might inflate the resulting association statistics, by explicitly modelling these confounders using linear hierarchical mixed models¹³⁵. However, hypothesis testing the very large number of possible correlations of analytes between different molecular layers would require unattainable sample sizes after multiple testing correction. Therefore, the hypothesis search space is often constrained by previous biological information (for example, the distance from the epigenetic marker or accessible regulatory element to the expressed gene). These approaches can be used to identify, for example, cell-type-specific allelic effects of genetic variation on gene expression (expression quantitative trait loci (eQTLs)) in differentiating induced pluripotent stem cells (iPS cells)¹³⁶ or the effect of genetic perturbations on gene expression using multiplexed CRISPR technologies to identify *cis*-acting enhancer–gene pairs¹³⁷. Local integration approaches can also be used to facilitate the inference of gene regulatory networks (GRNs). Classic GRN inference approaches, using only scRNA-seq data, identify sets of co-expressed genes enriched with particular upstream regulatory motifs¹³⁸. However, this approach is prone to false-positive gene associations as the presence of these motifs does not necessarily guarantee the activity of the associated TF. Using methods for simultaneous analysis of open chromatin and gene expression with sequencing, it has been shown that chromatin accessibility at regulatory domains precedes gene expression³⁸ and may thus outperform methods for GRN inference from scRNA-seq alone.

In contrast to local approaches, global integration strategies aim to identify larger-scale patterns (hundreds to thousands of features) of covariation across modalities to identify global shifts in the cellular state in an unsupervised way. To this end, linear matrix decomposition approaches such as principal component analysis (PCA), canonical correlation analysis (CCA)¹³⁹ and non-negative matrix factorization (NMF)^{140,141} have proven successful to identify latent factors underlying cellular heterogeneity. For example, MOFA+¹⁴² extends on this principle and builds on the Bayesian Group Factor Analysis framework to jointly model variation across spatial and temporal covariates and employs a sparse hierarchical prior architecture that handles both shared sources of variation and private variation that is particular to individual modalities. This approach was used to integrate the different modalities in scNMT-seq data to reveal lineage-specific enhancers during embryo development⁶⁸. Although the linearity of these approaches makes them inherently robust and results in readily interpretable factors of variation, they suffer from a lack of explanatory power when dealing with non-linearities that are commonly found in complex biological systems. Nonlinear multi-view neural network architectures^{143,144} such as multimodal autoencoders have been proposed to tackle this limitation. These neural networks aim to learn simple (low-dimensional) representations of the data, also known as an encoding, from complex (high-dimensional) input datasets. Instead of learning individual representations for each modality, these representations can be forced to be shared across

different data modalities, thus forcing the neural network to implicitly learn common links between them. Several alternative architectures have been proposed that optimize for different criteria such as robustness to dropouts and batch effects, disentanglement of the learned latent factors for improved interpretability and cross-modal translation for missing modality imputation^{145,146}. For further details

we refer the reader to a recent overview of proposed deep learning approaches¹⁴⁷.

Data integration for spatial multi-omics

Spatial mono-omics and multi-omics technologies also benefit from various data integration strategies. Before applying

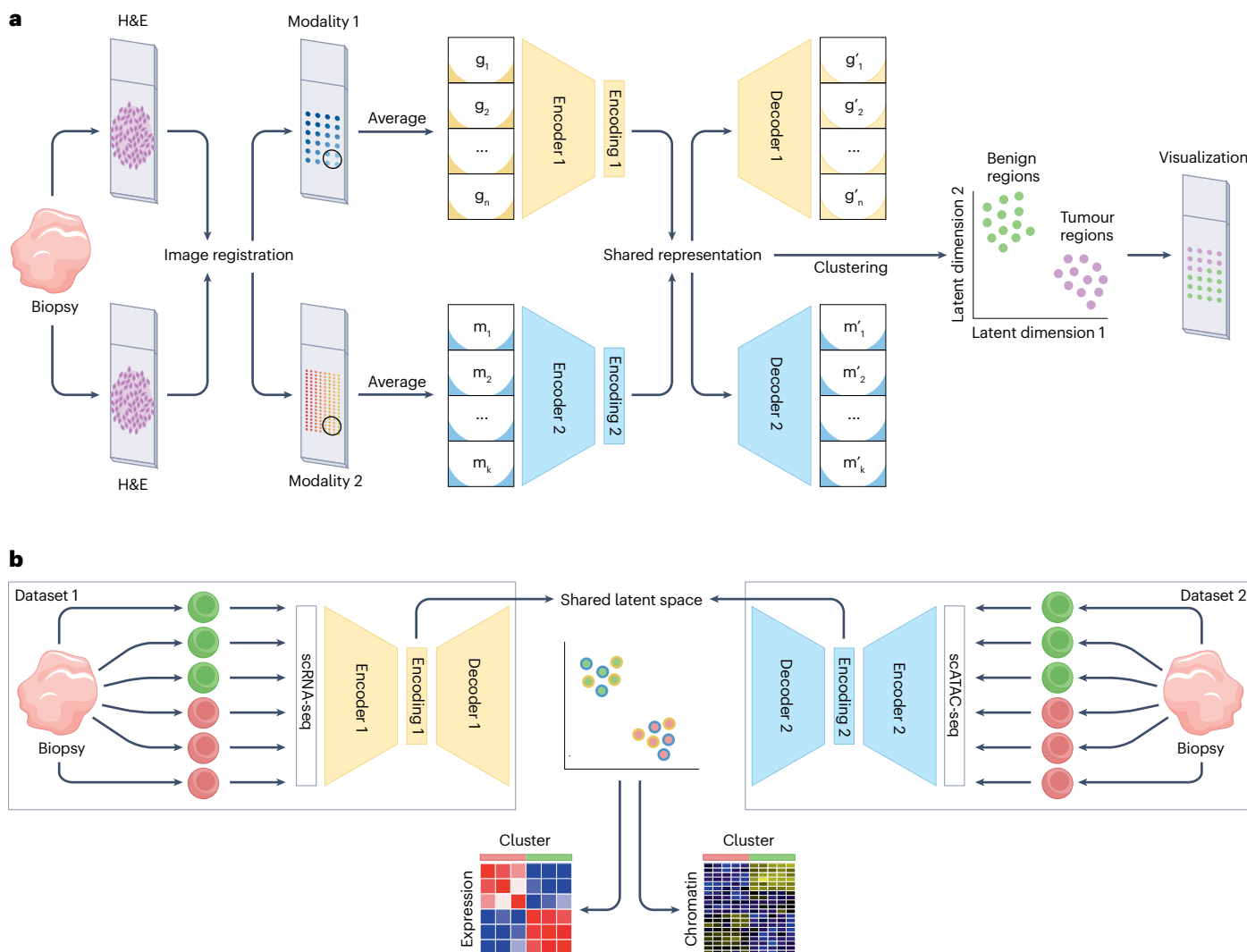


Fig. 6 | Data integration strategies. Examples of different scenarios in which various types of data integration strategies can be used. **a**, Vertical integration strategies aim to integrate information from paired molecular layers to obtain holistic representations of biological systems, at the single-cell level or at the tissue region level. Here, we illustrate an example of a spatial multi-omics experiment in which mirrored tissue slices have been assayed by two different spatial modalities (yellow and blue). To integrate both data sources, haematoxylin and eosin (H&E) staining images of each modality are first registered to account for deformations during sample preparation. Subsequently, to account for differences in resolutions across modalities, data points are averaged in windows of a predetermined region size¹⁵⁴. For every region the averaged regional profiles can be used as paired inputs for either linear or nonlinear vertical integration approaches. In this example, we illustrate integration through the use of a multi-view autoencoder neural network¹⁴³. Each modality is used as input into a dedicated encoder–decoder network that learns a shared data representation,

effectively integrating both data modalities. This shared representation can be used for downstream analysis and/or visualization. The terms g_i and m_i correspond to the measurements of region i for modality 1 and modality 2, respectively, with g'_i and m'_i being the molecular profiles reconstructed from the shared representation by the decoder networks. **b**, Integration of datasets from different experiments with independent observations of individual cells and non-overlapping molecular features is the hardest integration problem and requires diagonal integration approaches. Here, we illustrate this problem for the integration of independently acquired single-cell RNA sequencing (scRNA-seq) and single-cell assay for transposase-accessible chromatin (scATAC-seq) datasets through the use of autoencoder neural networks with a probabilistic coupling to map the different data modalities to a shared latent space¹⁴⁴. Although single-cell resolution is lost in this coupling, clusters obtained in this shared latent space can be used to ascertain correlations between molecular layers, discover multimodal biomarkers and/or translate between the different modalities.

Glossary

3C or Hi-C

A class of methods uncovering the higher-order chromosome conformation and chromatin interactions in the nucleus.

Autoencoders

A class of architectures of neural networks that take a highly dimensional input and encode into a low-dimensional representation via an encoder network. These networks are trained in a self-supervised way by reconstructing the original input from the encoding using a decoder network.

Bisulfite sequencing

A method in which bisulfite treatment of DNA before sequencing converts unmethylated cytosines to uracil, while 5-methylcytosines are protected from conversion. This method hence discloses sites in the genome where DNA methylation is found.

Cell plasticity

The ability of a cell to remodel its epigenomic, transcriptomic and proteomic landscape, leading to new phenotypic features.

Cell state

A specific stable or unstable functional condition of a cell belonging to a particular cell type.

Cell type

A categorization of cells by specific morphological and/or phenotypic characteristics.

Chromatin accessibility

The structures of DNA wrapped around histone octamer proteins (nucleosomes) that make up chromosomes. Chromatin can be highly condensed with DNA inaccessible or open with DNA accessible.

Displacement preamplification followed by PCR

(DA-PCR). A class of methods for whole-genome amplification of single-cell DNA using multiple rounds of strand displacement amplification, generating amplicons with PCR-handles at both ends, enabling molecules to be further amplified by PCR.

DNA methylation

The deposition of methyl groups onto DNA. In mammals, methylation is predominantly found at cytosines as 5-methylcytosine, and mostly in a cytosine–guanine dinucleotide (CpG) context.

Epigenome

The ensemble of modifications to DNA and DNA-associated proteins that signal and regulate gene expression and other DNA-related processes. Examples include DNA methylation, histone post-translational modifications and chromatin remodelling proteins.

Exome sequencing

A targeted next-generation sequencing approach that enables sequencing the coding exons, which can be supplemented with flanking intronic sequences as well as (part of) the gene regulatory sequences, approximating 1.1% or more of the human genome.

FFPE

Formalin-fixed paraffin-embedded (FFPE) tissues are preserved for long-term archival storage by formalin fixation followed by embedding in paraffin wax blocks.

Genome

The genetic blueprint or DNA of an organism established at fertilization, which for humans normally consists of 23 pairs of chromosomes contained within the nucleus, plus the mitochondrial DNA molecules present at multiple copies in the mitochondria of the cell.

Hierarchical mixed models

Statistical regression models that model nested hierarchies in the data taking into account complex covariate data structures.

Human Cell Atlas

The Human Cell Atlas (HCA) aims to create a reference map of all human cells, providing a basis for understanding human health and for diagnosing, monitoring and treating diseases.

Linear matrix decomposition

Numerical analysis algorithms that factorize observations into a product of smaller matrices. These matrices usually represent factors or components of variation that are easier to interpret (such as biological processes) and their presence within the observations (cells).

Metabolome

The ensemble of metabolites present within a cell or tissue.

Multiple displacement amplification

(MDA). An isothermal DNA amplification method that applies DNA polymerase(s) with strand displacement activity.

Neural networks

A class of algorithms that use densely connected networks of artificial neurons that are non-linearly activated given a combination of input values from connected input neurons, mimicking the human brain.

OCT compound

Optimal cutting temperature (OCT) compound is a standard type of sample-embedding medium used to embed fresh-frozen tissue samples for optimal storage and cryosectioning.

Phenome

The set of traits or characteristics expressed by an organism or cell.

Proteome

The ensemble of proteins translated from transcribed genes contained within the genome, which can be highly variable among different cell types.

Smart-seq2

Switching mechanism at the end of the 5' end of the RNA transcript (Smart)-based method for single-cell cDNA generation, amplification and sequencing library conversion.

Tagmentation

A method in which double-stranded DNA is cleaved and tagged with adapter sequences in a single step by using a transposase complex loaded with these adapter sequences.

Template switching

A method in which Moloney murine leukaemia virus (MMLV)-type reverse transcriptases add non-templated nucleotides (CCC) to first-strand cDNA near the 5' end of the transcript, enabling annealing of an rGrGrG-containing template-switching oligonucleotide and the reverse transcriptase to switch templates and copy the template-switching oligonucleotide sequence to the cDNA.

Transcriptome

The ensemble of transcripts or RNA molecules transcribed from genes contained within the genome, which can be highly variable among different cell types.

Unique molecular identifier

Short sequences that barcode each molecule individually and are added as tags to DNA fragments in next-generation sequencing-based approaches to identify molecules of interest with increased accuracy.

vertical integration, horizontal integration algorithms can be used to correct for technical variability between different sections of the same or different samples, and can also be expanded to fully

utilize spatial correlations between sections¹⁴⁸. Additionally, for spatial omics technologies that do not achieve single-cell resolution, deconvolution methods can be used to integrate non-spatial

single-cell data to identify the cellular constituents of a particular tissue region^{149–151}.

Vertical integration of non-spatial single-cell multi-omics data followed by spatial deconvolution using the RNA modality allows other molecular modalities to be indirectly spatially mapped. This approach was used, for example, by Foster et al., using ArchR¹⁵² for scRNA-seq and scATAC-seq integration, to identify spatial fibroblast epigenome patterns during tissue repair¹⁵³. Spatial multi-omics technologies also allow for vertical integration, although few currently available technologies allow for true multimodal assaying at single-cell resolution on the same tissue section. Hence, generating spatial multi-omics data often still relies on the analysis of adjacent tissue sections and is thus limited to studying correlations across modalities of tissue regions (Fig. 6a). Linear approaches are suitable for the inference of these correlations, although special care needs to be taken in the statistical analysis as spatial autocorrelation can violate independence assumptions of the observed datapoints^{154,155}. Although more spatial multi-omics technologies are emerging, spatial mono-omics assays can often also be combined with simple histological stains, such as H&E staining, allowing for the integration of cellular morphology and gene expression data. Nonlinear neural networks have been proposed to map discrete spot-based expression profiles onto high-resolution morphology images¹⁵⁶ or to learn a joint cell morphology-plus-expression representation for the identification of novel cellular subpopulations missed by individual modalities¹⁵⁷. As more spatial multi-omics technologies become available that can characterize many analytes at single-cell resolution, we expect rapid developments of vertical integration strategies based on adaptations of currently available non-spatial single-cell methodologies. By leveraging spatial information linking a cellular state to its respective micro-environments and macro-environments (for example, through the use of graph neural networks), more fine-grained multimodal representations of cellular state should be obtainable.

Finally, approaches for the harder problem of diagonal integration of unpaired spatial and non-spatial data modalities have also been proposed using autoencoder neural network architectures with a probabilistic coupling between the different data modalities. For these approaches, however, previous knowledge is generally required to constrain the resulting shared representations to biologically accurate alignments¹⁴⁴.

Current challenges in data integration

Despite extensive research, several challenges remain for computational data integration. These strategies make implicit assumptions about the expected similarity of cellular states captured across different experiments and/or modalities. In the case of horizontal data integration across different batches, this can lead to overcorrection of true biological variation¹⁵⁸, especially when studying subtle shifts in cellular state in different experimental conditions. This problem is further exacerbated in vertical integration strategies when correspondence between features is not immediately obvious and/or when non-linearities between the different layers might be biologically relevant¹³². For example, gene expression changes have been shown to be foreshadowed by chromatin priming, as seen in cell cycling via SHARE-seq³⁸. Also, other biological phenomena such as alternative splicing and/or PTMs are likely to influence the correlation structure in cell state across the different molecular layers. Therefore, integration methods need to be developed that can not only identify the common anchors between the different samples and/or molecular layers but can also account for sample-specific and modality-specific variation in a

readily interpretable way. Such models have already been proposed in other fields of multi-domain learning but have not yet been adapted to cellular multi-omics datasets¹⁵⁹. The expansion of these models to integrate paired, unpaired and multimodal spatial assays will require robust data standards^{160,161} for scalable analysis of multimodal data in addition to well-established benchmarks to evaluate their efficacy.

Perspectives

In the decades to come, multi-omics at single-cell and spatial resolution will innovate further, leading to a more holistic understanding of cell biology. Advances can be anticipated on multiple fronts, including improvements in throughput, reduction of cost and the incorporation of more modalities in a single assay. Additionally, we expect improvements of sensitivity and specificity in the detection and characterization of each modality as part of multi-omic measurements. For example, on the genome level, full and error-free characterization of all genetic variants is still a challenge, which currently limits opportunities for comprehensive somatic mutation profiling at single-cell resolution and reconstructing phylogenetic cell lineages from naturally acquired mutations. Similarly, measurements of the epigenome are severely limited for co-detection of the range of epigenomic features co-regulating gene expression and other DNA-related processes. For instance, histone PTMs can currently be detected only a single or a few marks at a time; hence, these methods would benefit from drastic increments in the number of PTMs that can be co-detected, also together with other epigenomic features. The characterization of the transcriptome is often limited to poly(A) RNA rather than total RNA measurements; hence, it would be beneficial to include both coding and (small) non-coding RNAs as well as concurrent isoform detection. Proteome assays are still antibody-based and thus limited in how many proteins can be profiled simultaneously. Unbiased low-input methods such as mass-spectrometry-based approaches may circumvent this, but cannot currently be combined with assays for other molecular layers. This lack of multimodal integration also holds true for metabolome and lipidome assays. In addition to such improvements for characterizing modalities in multi-omics methods, we also anticipate the development of multimodal assays that incorporate entirely new modalities that currently remain uncharted, such as the epitranscriptome (consisting of base modifications to transcripts, some of which are known to affect gene expression)¹⁶². Furthermore, the field is likely to see a continued strong technological push for spatial multi-omics, avoiding the need for tissue dissociation and enabling concurrent multi-omic profiling of cell-intrinsic and -extrinsic molecular features defining cell types and states. Moreover, integration of phylogenetic cell lineages reconstructed from naturally acquired or artificially induced DNA mutations with other spatial or single-cell multi-omic information will transform our understanding of organismal development, cell migration routes and stem cell biology in health and disease. Finally, it will be necessary to develop methods that capture not only transient phenotypes but also ancestral states, apply multi-omic technologies to serial measurements in live cells¹⁶³, and computationally improve the accuracy of data extraction from each molecular layer, as well as undertake integrative analyses across modalities to unravel dependencies within and across different data sources.

To arrive at such holistic multi-omic profiling of single cells dissociated from tissue or at spatial resolution will require many challenges to be overcome in the following years. For a view on how challenges associated with each modality specifically could be tackled, we refer to previous reviews on single-cell and spatial mono-omic profiling^{11,86,164–168}.

Optimization of low-input bulk molecular analysis tools to single-cell and/or spatial resolution, like amplification-free long-read single-molecule sequencing approaches, will enable the incorporation of novel molecular readouts. In parallel, to enable integration of more modalities per assay will require innovation in the principles presented in Fig. 2, including expansion and combination of the different principles. Making various single-cell and spatial multi-omics assays commercially available will also make them more accessible and applicable for the wide research community, but we are already starting to see this evolution¹⁶⁹.

All these technological and computational developments will translate into a better understanding of development, organismal function and functional decline of organs with ageing, and will be key in unravelling the cellular pathogenesis of diseases, identifying more effective stratification strategies of disease processes, devising novel therapeutic modalities and guiding precision medicine.

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Author contributions

The authors contributed equally to all aspects of the article.

Competing interests

T.V. is co-inventor on licensed patents WO/2011/157846 (Methods for haplotyping single cells), WO/2014/053664 (High-throughput genotyping by sequencing low amounts of genetic

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